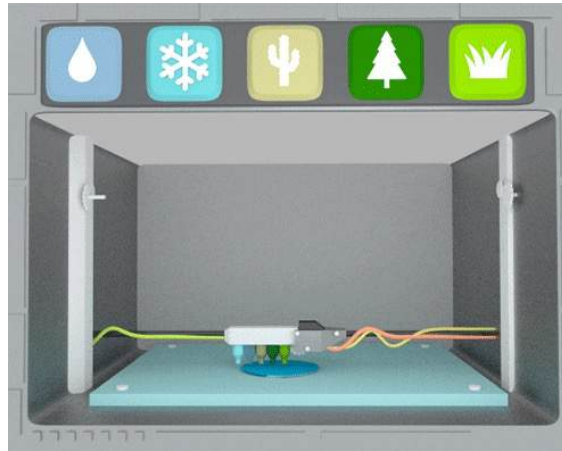




Source: 3D Printing World Environment Day GIF By General Electric

Embedding Sustainability in Design Optimization:
Which Mathematical Recipes?

Prof. Joseph Morlier

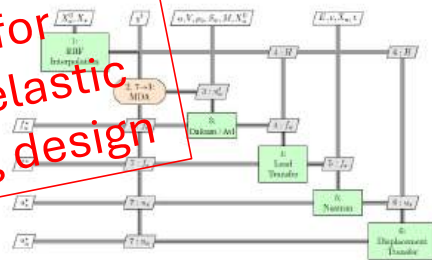


Thanks to all my
PhDs, MsCs and
co-authors
Edouard Duriez,
Gustavo Asai, Paul
Saves, Luis Yepes
Llorente, Thomas
Bellier

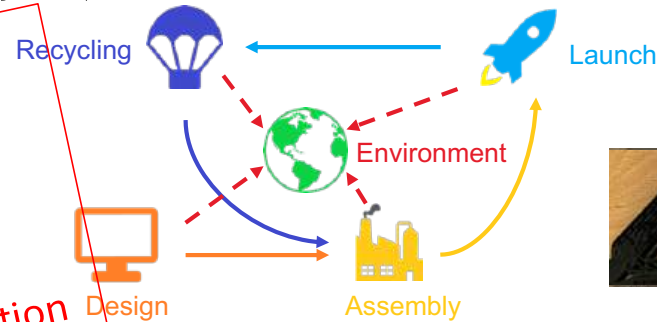
<https://pagespro.isae-supaero.fr/joseph-morlier/>

- Prof in Structural and Multidisciplinary Optimization
- 6 PhDs, 3 MsCs

MDO for Aeroelastic Wing design

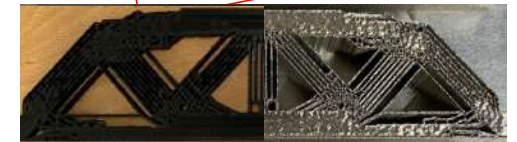


MDO for Aerospace systems Including LCA
:= EcoOptimization

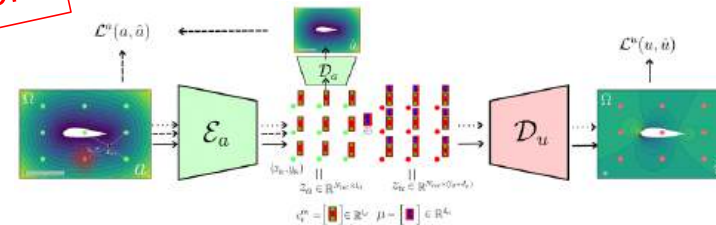


About Me?

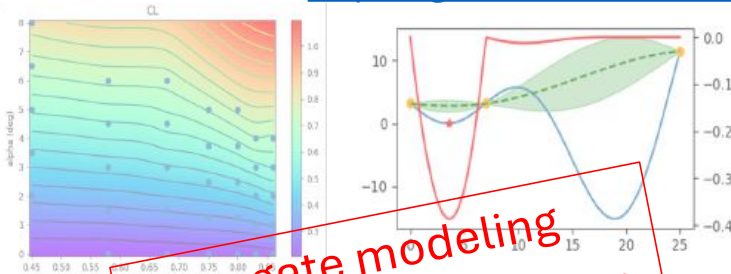
Digital fabrication



AI4E



<https://github.com/SMTorg/SMT>



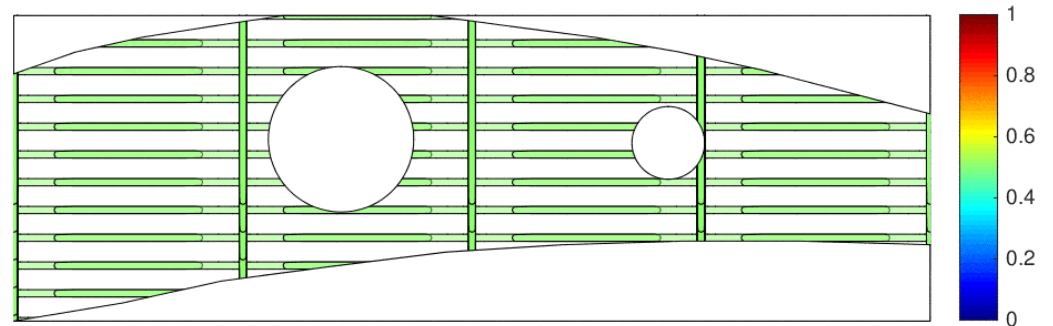
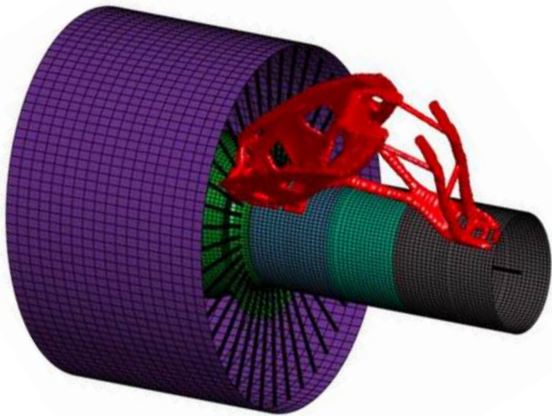
Surrogate modeling
Bayesian Optimization



Séminaire Pluridisciplinaire d'Optimisation de Toulouse

Recap first

- What is TOPOLOGY optimization ?



History

- Homogenization of Microstructures was introduced by mathematics in the 1970s.
- First paper by Martin Bendsoe (Technical University of Denmark) and Noboru Kikuchi (University of Michigan) in 1988

A topology optimisation problem can be written in the general form of an optimization problem as

$$\min_{\rho} F = F(\mathbf{u}(\rho), \rho) = \int_{\Omega} f(\mathbf{u}(\rho), \rho) dV$$

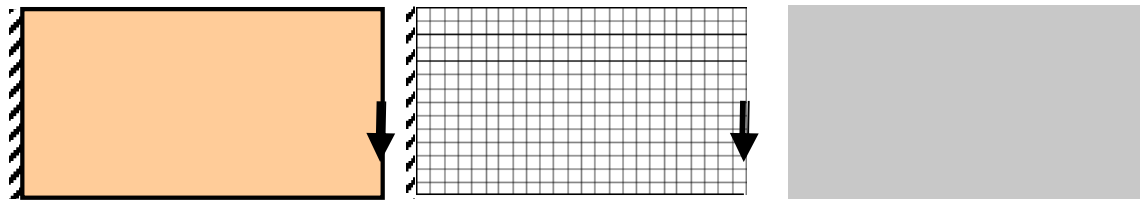
subject to

- $\rho \in \{0, 1\}$
- $G_0(\rho) = \int_{\Omega} \rho(\mathbf{u}) dV - V_0 \leq 0$
- $G_j(\mathbf{u}(\rho), \rho) \leq 0$ with $j = 1, \dots, m$

$$\left. \begin{array}{l} \min_{\mathbf{x}}: c(\mathbf{x}) = \mathbf{U}^T \mathbf{K} \mathbf{U} = \sum_{e=1}^N (x_e)^p \mathbf{u}_e^T \mathbf{k}_0 \mathbf{u}_e \\ \text{subject to: } \frac{V(\mathbf{x})}{V_0} = f \\ \quad : \mathbf{K} \mathbf{U} = \mathbf{F} \\ \quad : 0 < \mathbf{x}_{\min} \leq \mathbf{x} \leq 1 \end{array} \right\},$$

SIMP in action

Solid Isotropic Material with Penalization



Min Compliance
 $v = 0.5v_0$

$0 < \rho_e \leq 1$: 'PseudoDensity'

Where do we add
 holes?

$$\begin{aligned} & \text{Min}_{\rho_e} \text{ Compliance} \\ & \sum \rho_e v_e = 0.5v_0 \end{aligned}$$

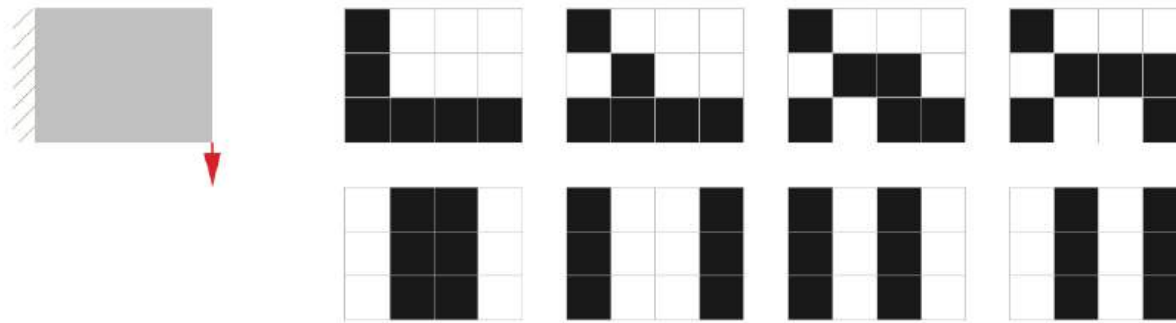
$$\left. \begin{aligned} \min_{\mathbf{x}} : & \quad c(\mathbf{x}) = \mathbf{U}^T \mathbf{K} \mathbf{U} = \sum_{e=1}^N (x_e)^p \mathbf{u}_e^T \mathbf{k}_0 \mathbf{u}_e \\ \text{subject to :} & \quad \frac{V(\mathbf{x})}{V_0} = f \\ & \quad : \quad \mathbf{K} \mathbf{U} = \mathbf{F} \\ & \quad : \quad 0 < \mathbf{x}_{\min} \leq \mathbf{x} \leq 1 \end{aligned} \right\},$$

Pixels?



Pixels

- Finding a solution by checking all the possible combinations IS impossible since the number of topologies nT increases exponentially with the number of finite elements n
- $nT = 2^n$,



The legal (top) and some illegal (bottom) topologies with 4 by 3 elements

Division into elements (pixels or voxels) and binary decision for each
example 10,000 elements --> 210,000 possible configurations!

State-of-Art – SIMP^{1,2}

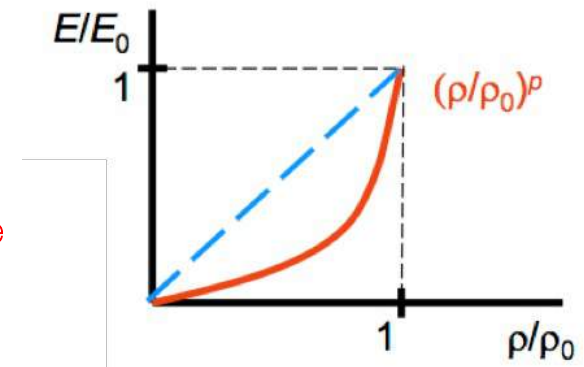
Predict an optimal distribution of the material in a design space for given boundary conditions and design constraints

Density-based topt

$$E_e = E_{min} + x_e^p (E_0 - E_{min})$$

P chosen as 3, or iterative

$$c(x) = U^T K U = \sum_{e=1}^N E_e (x_e)^p u_e^T k_0 u_e$$



1 - Bendsøe MP (1989) Optimal shape design as a material distribution problem.

2 - Zhou M, Rozvany G (1991) The COC algorithm, part II: topological, geometrical and generalized shape optimization.

3 - Andreassen E, Clausen A, Schevenels M, Lazarov B, Sigmund O (2010) Efficient topology optimization in MATLAB using 88 lines of code

Compliance minimization

- Compliance is the opposite of stiffness

$$C = \mathbf{f}^T \mathbf{u} = \mathbf{u}^T K \mathbf{u}$$

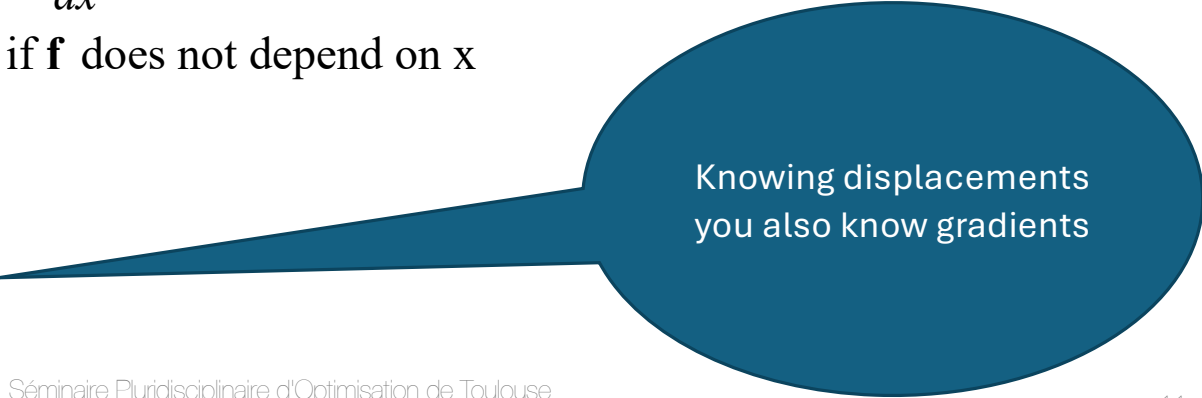
- Inexpensive derivatives (use chain rule)

$$\frac{dC}{dx} = 2\mathbf{u}^T K \frac{d\mathbf{u}}{dx} + \mathbf{u}^T \frac{dK}{dx} \mathbf{u}$$

But since $K\mathbf{u} = \mathbf{f}$ if $\mathbf{f}$ does not depend on x

$$K \frac{d\mathbf{u}}{dx} = -\frac{dK}{dx} \mathbf{u}$$

$$\frac{dC}{dx} = -\mathbf{u}^T \frac{dK}{dx} \mathbf{u}$$



Knowing displacements
you also know gradients

Density design variables

- Recall $\frac{dC}{dx} = -\mathbf{u}^T \frac{dK}{dx} \mathbf{u}$

- For density variables

$$\frac{dC}{d\rho^e} \propto -\mathbf{u}^T \rho^{p-1} K^e \mathbf{u}$$

- Want to increase density of elements with high strain energy and vice versa
- To minimize compliance for given weight can use an optimality criterion method.

And for other responses?

$$O = f(x, U)$$

$$\frac{\partial O}{\partial x} = \frac{\partial f^T}{\partial U} \frac{\partial U}{\partial x}$$

$$KU = F$$

$$\frac{\partial K}{\partial x} U + K \frac{\partial U}{\partial x} = 0$$

$$\frac{\partial O}{\partial x} = \frac{\partial f^T}{\partial U} \frac{\partial U}{\partial x} = -\frac{\partial f^T}{\partial U} K^{-1} \frac{\partial K}{\partial x} U = -\frac{\partial f^T}{\partial U} \delta$$

$$K\lambda = \frac{\partial f}{\partial U} \text{ Adjoint Method}$$

$$K\delta = \frac{\partial K}{\partial x} U \text{ Direct Method}$$



Either one solution per
response
Either one solution per design
variables
That's why Compliance!

Au programme

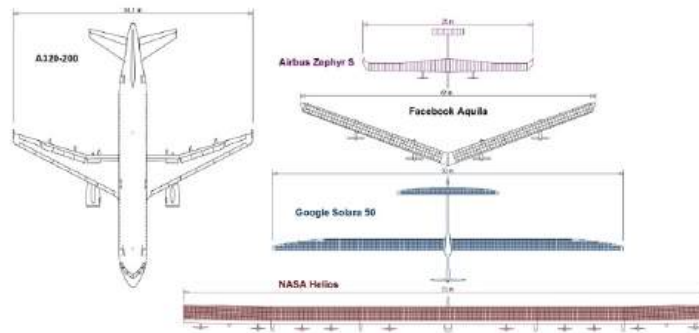
1. Problem $\mathcal{P}_1$, where the topology is fixed (truss) or parametric (wing),
2. Problem $\mathcal{P}_2$, where the topology is free.
3. On some coupled problems $\mathcal{P}_3$
4. Conclusions

Au programme

- 1. Problem $\mathcal{P}_1$, where the topology is fixed (truss) or imposed (wing),**
2. Problem $\mathcal{P}_2$, where the topology is free.
3. On some coupled problems $\mathcal{P}_3$
4. Conclusions

$$\mathcal{P}_1$$

- Continuous relaxation *versus* Variational Autoencoders (VAEs)



ECO Hale

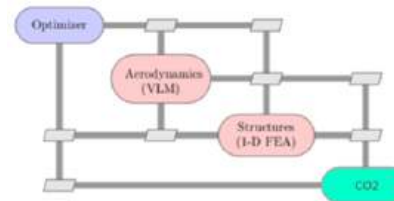
Assets: Flexible, repositionable, permanent coverage, cheaper, lower environmental impact?

Derive aerostructures MDO tools to treat a HALE pseudo satellite Design problem

scientific reports

OPEN **CO₂ footprint minimization of solar-powered HALE using MDO and eco-material selection**

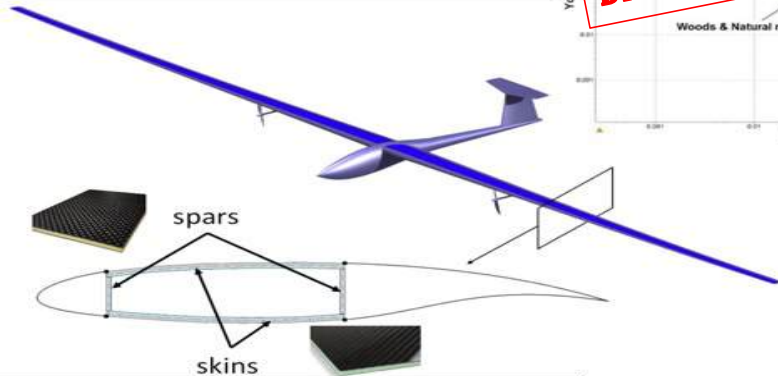
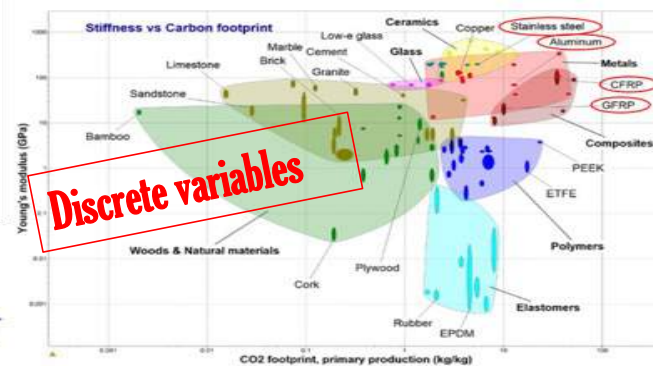
Edouard Duriez^{1,2}, Victor Manuel Guadaño Martín^{2,3} & Joseph Morlier^{2,4}



Thicknesses, twists, material, geometry

CO₂ footprint

Available power, stress, flight



Minimize CO₂
w.r.t. thicknesses, twist, geometry,
materials database CES EDUPACK (CFRP-3, GFRP, ALU,...)
Subject to Available solar power,
stress, buckling, flight

aire Pluridisciplinaire d'Optimisation de Toulouse

ECO MATERIAL SELECTION

Turn discrete problem of material selection into continuous inspired from SIMP

Zuo, W., & Saitou, K. (2016). Multi-material topology optimization using ordered SIMP interpolation. Structural and Multidisciplinary Optimization, 55(2), 477-491. doi:10.1007/s00158-016-1513-3

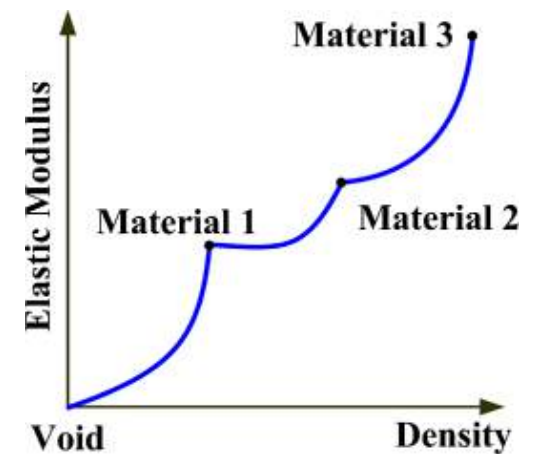
- Access material properties through density (only one material design variable) :

$E(\rho)$, $G(\rho)$, $\sigma_{\text{yield}}(\rho)$, $\text{CO}_2(\rho)$ (production CO2 footprint)

- Example for young modulus : Between two materials i and $i+1$,

$$E_e(\rho_e) = A_E * \rho_e^p + B_E, \text{ with } A_E = \frac{E_i - E_{i+1}}{\rho_i^p - \rho_{i+1}^p} \text{ and } B_E = E_i - A_E * \rho_i^p$$

- Power p added during optimisation so that only real materials are optimal
- Inversed curvature for CO2 as smaller CO2 is more advantageous



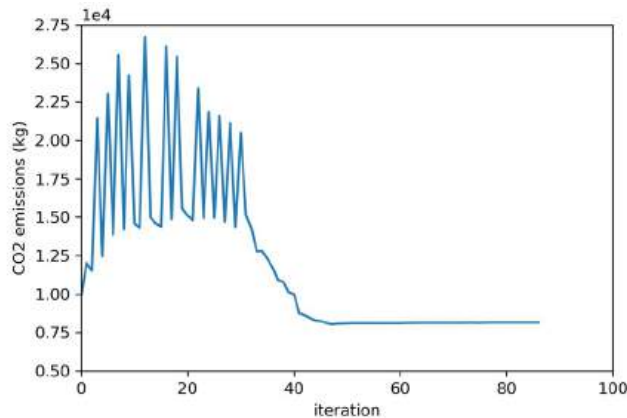
Our design variables

wing **topology** and **materials** !

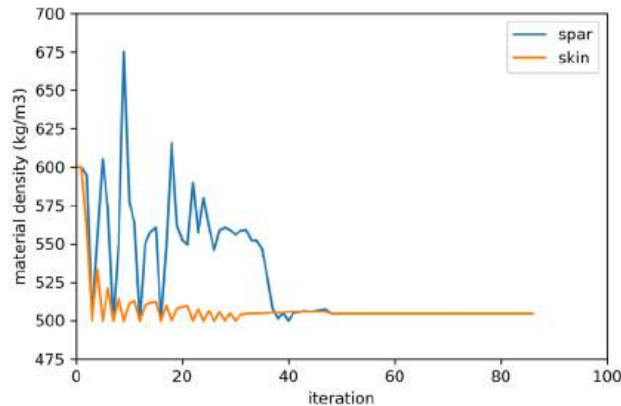
Variable	Lb	Ub	X0 (multistart)	unit
Skin thickness	0,0001	0,1	0,01-0,001	m
Spar thickness	0,0001	0,1	0,01-0,001	m
Wing span	1	1000	50	m
Wing chord	1	500	2	m
Wing taper	0,01	0,99	0,1	-
Wing thickness over chord ratio	0,01	0,20	0,1	-
Twist	-15	15	5	°
Density	50	8200	50-8200	Kg/m3

CO2 footprint minimization

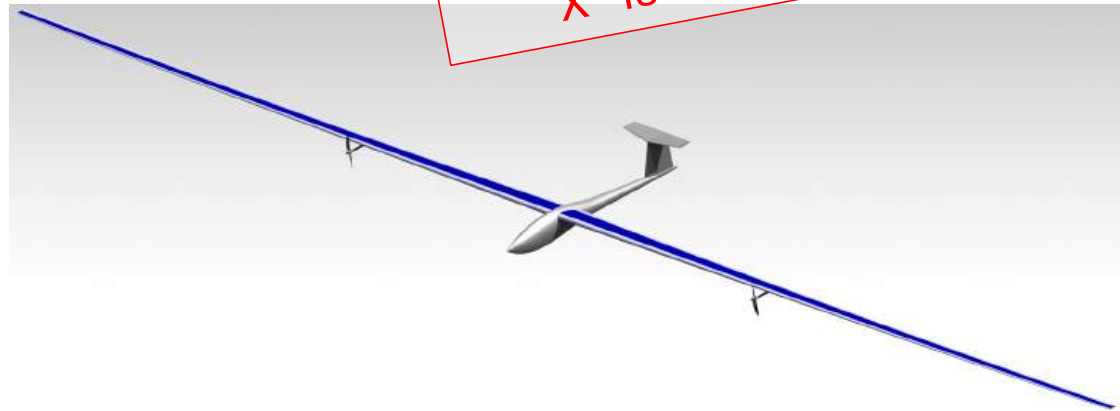
(a) Objective function: total CO2 emitted:



(b) Material density for skins and spars:



Convergence graphs



CAD model of the optimal HALE obtained

A slight increase in the total weight of the drone leads to an increase in the weight of the battery and the solar panel in order to propel a heavier drone,

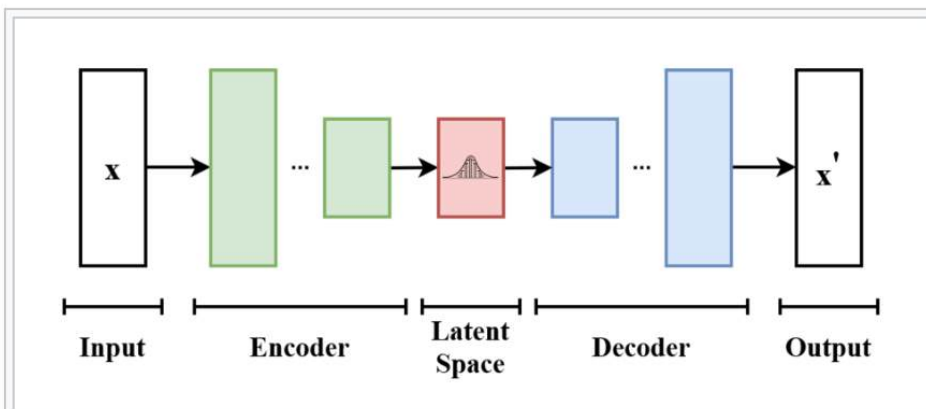
But also to: an increase in the weight of the wing structure that induces a more important lift to compensate → increase in the overall weight of the drone.

→ "snowball" effect.

Few years later

The main problem: introducing *discrete* material database in a *continuous* gradient based optimization

the environmental choices most often rely on database of discrete properties (Ansys Granta, Ecoinvent etc...).



The basic scheme of a variational autoencoder. The model receives x as input. The encoder compresses it into the latent space. The decoder receives as input the information sampled from the latent space and produces x' as similar as possible to x .

HOW?

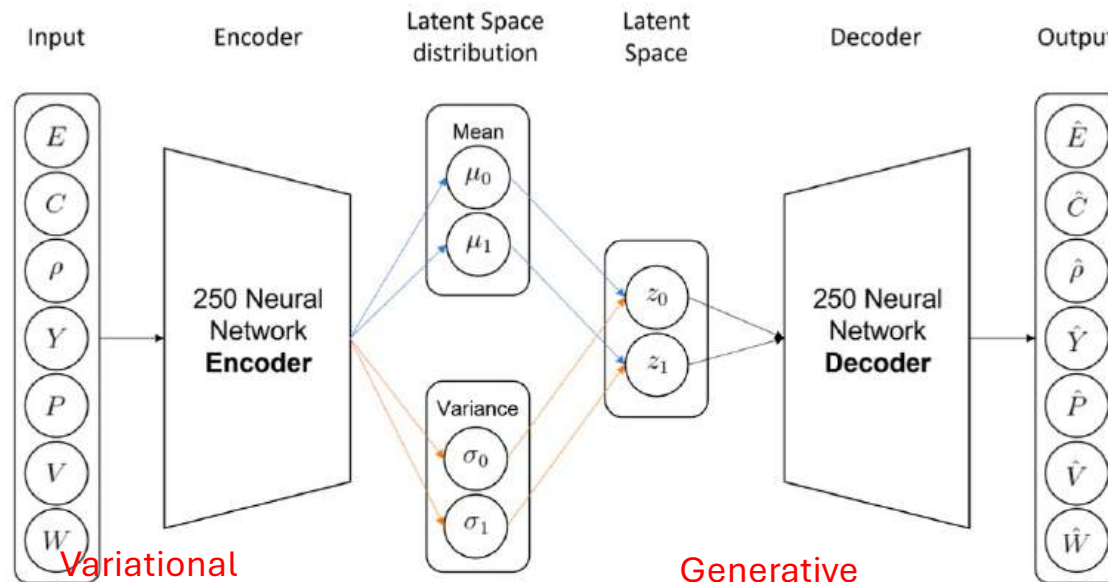
Variational AutoEncoder (VAE):
class of probabilistic models that find latent, low-dimensional representations of data.

Kingma, D. P., & Welling, M. "Auto-Encoding Variational Bayes", 2013.

From catalog to *differentiable* material representation

Table 1 A curated subset of materials and their training

Material	Class	E [Pa]	Cost C [\$/kg]
A286 iron	Steel	2.01E+11	5.18E+00
AISI 304	Steel	1.90E+11	2.40E+00
Gray cast iron	Steel	6.62E+10	6.48E-01
3003-H16	Al alloy	6.90E+10	2.18E+00
5052-O	Al alloy	7.00E+10	2.23E+00
7050-T7651	Al alloy	7.20E+10	2.33E+00
Acrylic	Plastic	3.00E+09	2.80E+00
ABS	Plastic	2.00E+09	2.91E+00
PE HD	Plastic	1.07E+09	2.21E+00



Variational
posterior
as an Encoder

Generative
model
as a Decoder

Framework: PyTorch VAE

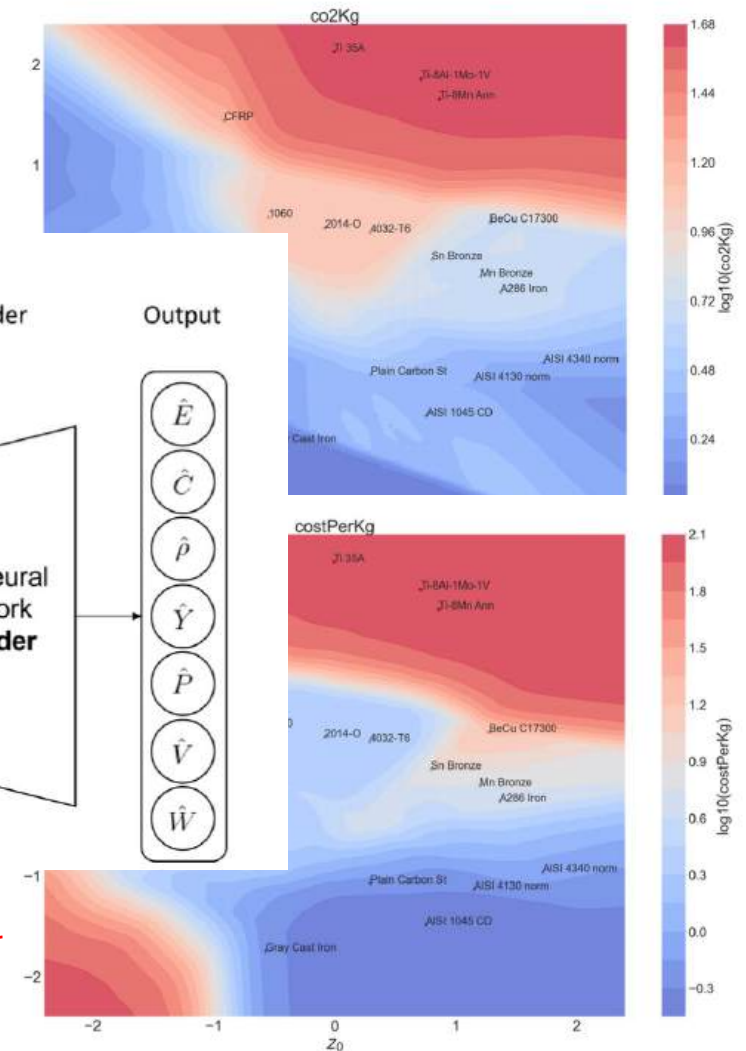
Network architecture: Fully connected with 250 neurons in encoder and decoder

Latent space: 2D (z_0, z_1)

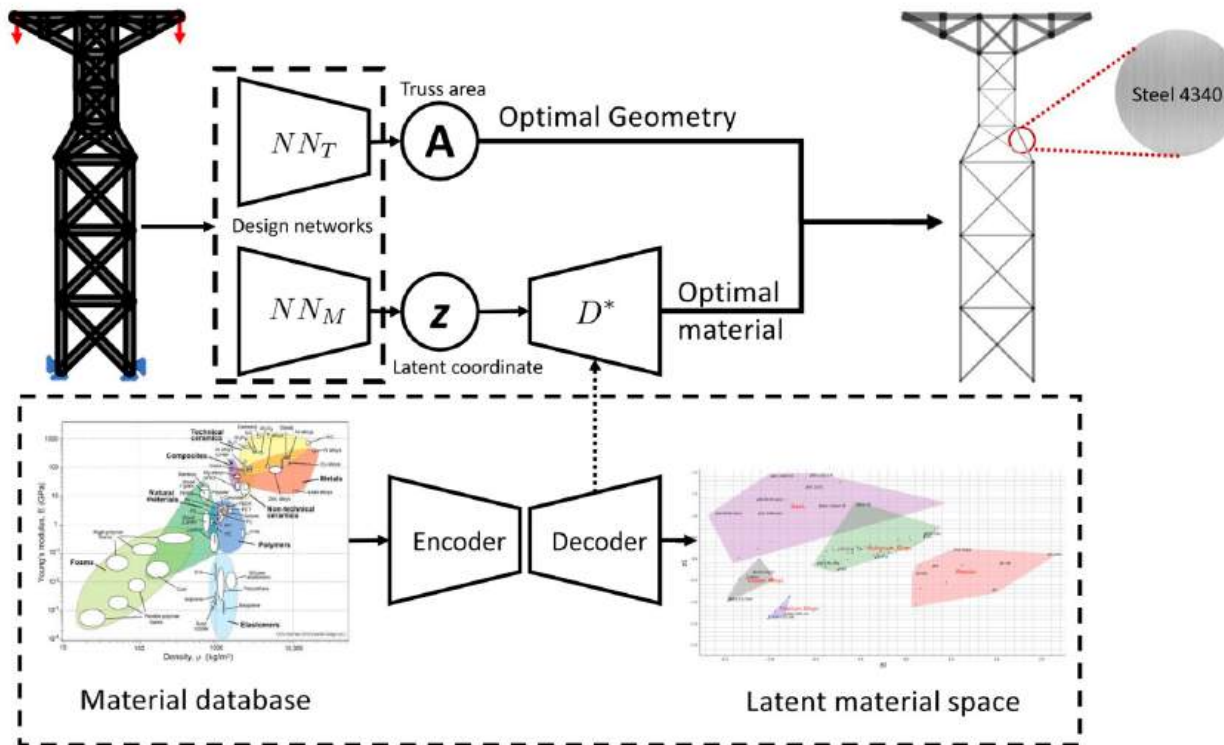
Inputs normalized to $[0, 1]$; outputs rescaled post-decoding

Training time: 3 minutes on a CPU (50,000 iterations)

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Everything is continuous here !



Chandrasekhar, A., Sridhara, S., & Suresh, K. (2022). Integrating material selection with design optimization via neural networks. *Engineering with Computers*, 38(5), 4715-4730.

Yepes Llorente, L., Morlier, J., Sridhara, S., & Suresh, K. (2024). A hybrid machine learning and evolutionary approach to material selection and design optimization for eco-friendly structures. *Structural and Multidisciplinary Optimization*, 67(5), 69.

LCA & eco selection

- Material
- Process
- from cradle to grave
- ...

Water
withdrawal



Generation of
waste



Use recycled:
Fibers
Resin
Metals
Reuse & Repair

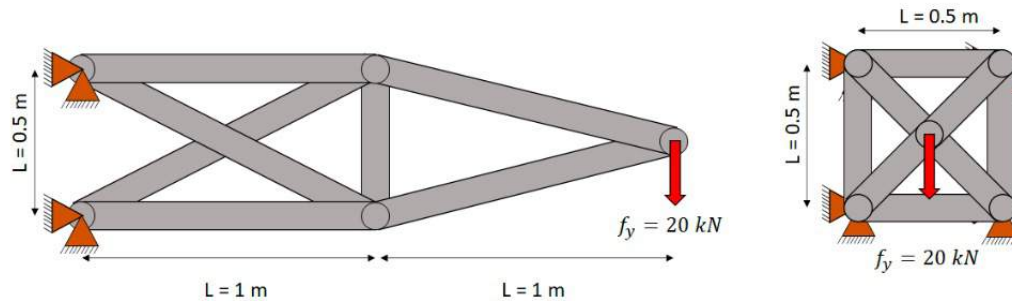
Carbon
Footprint



Energy
requirement



Environmentally friendly 3D truss

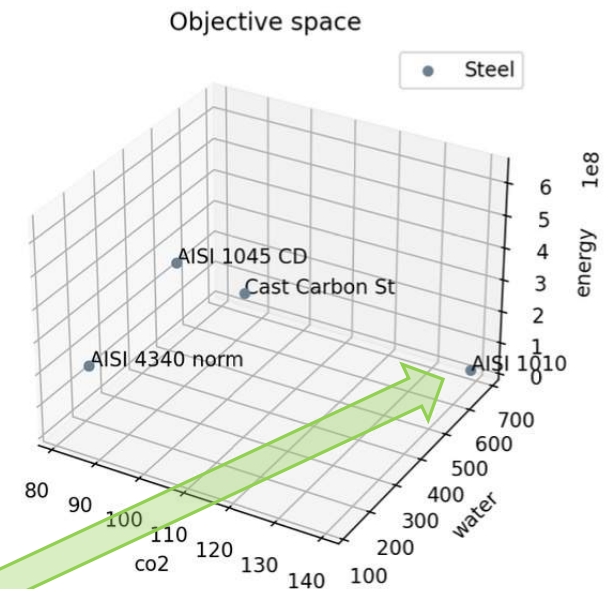


Being P_{min} , W_{min} and V_{min} the minimum value of CO_2 , water and energy achievable respectively, the index is defined as:

$$H = \frac{1}{3} \frac{P - P_{min}}{P_{min}} + \frac{1}{3} \frac{W - W_{min}}{W_{min}} + \frac{1}{3} \frac{V - V_{min}}{V_{min}} \quad (7)$$

In such a way that the perfect material with minimum consumption would yield $H = 0$. To take into account the cost of the structure as well, both values were calculated for each solution and presented in Table 10.

Material	Cost	Env. Consumption
AISI 1045 CD	18.63 \$	9.60 [-]
Cast Carbon St	28.68 \$	7.85 [-]
AISI 4340 norm	31.91 \$	2.28 [-]
AISI 1010	24.04 \$	1.73 [-]



Au programme

1. Problem $\mathcal{P}_1$, where the topology is fixed (truss) or imposed (wing),
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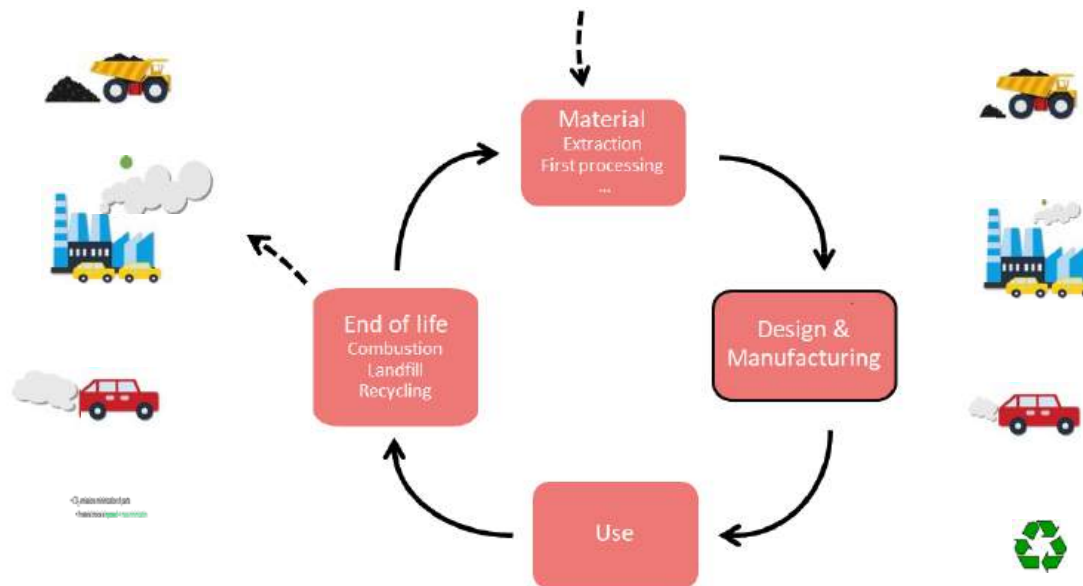
$$\mathcal{P}_2$$

$$\begin{array}{ll} \min_{m,p,\mathcal{D}} & CO_2^{\text{tot}}(\mathcal{D}, m, p) \\ \text{s.t.} & \delta \leq \delta_{\max} \\ & \{m, p\} \in \Phi_{\text{material}} \times \Phi_{\text{process}} \end{array}$$

- Topology, material and process.

Hypothesis 1

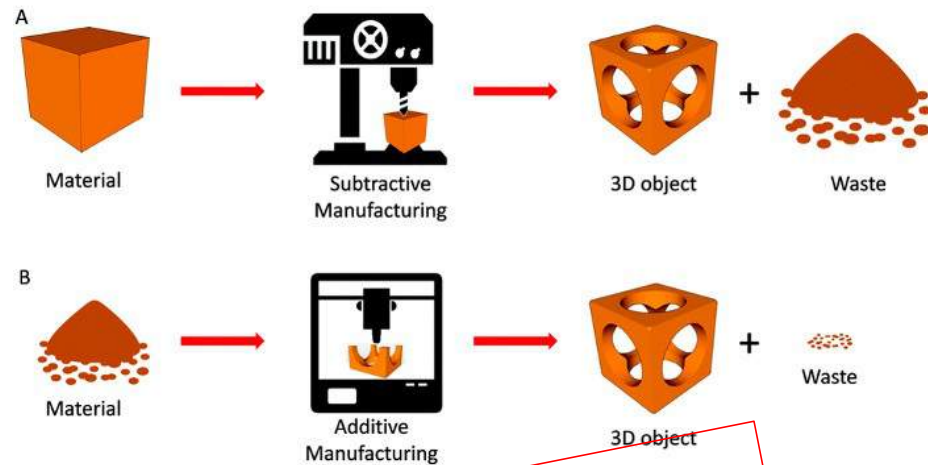
- CO_2 emissions minimization of parts
 - If material choice is **imposed** => mass minimization



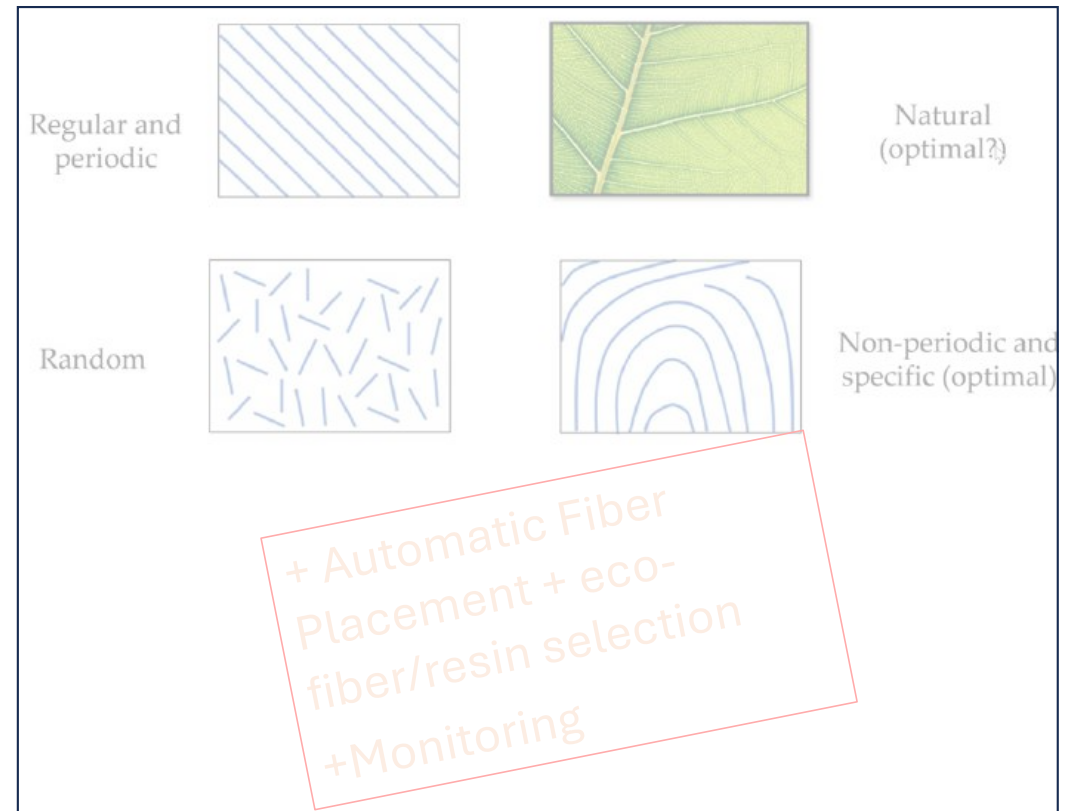
Vehicle life
cycle

Process is AM, but WHY?

<https://dfam.substack.com/p/dfam-education-in-2022>

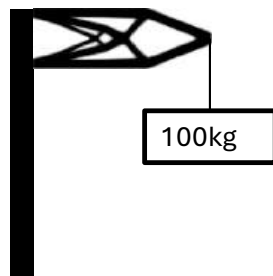
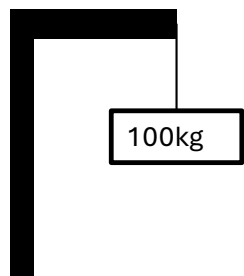


+Near 100% material utilization +Recyclability, Buy to fly ratio
+LCA of 3D printing machine
+Monitoring

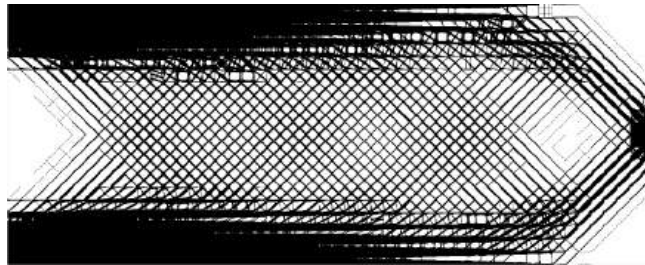


Mass minimization of parts

- Redesign through topology optimization
=> same performance but lower mass

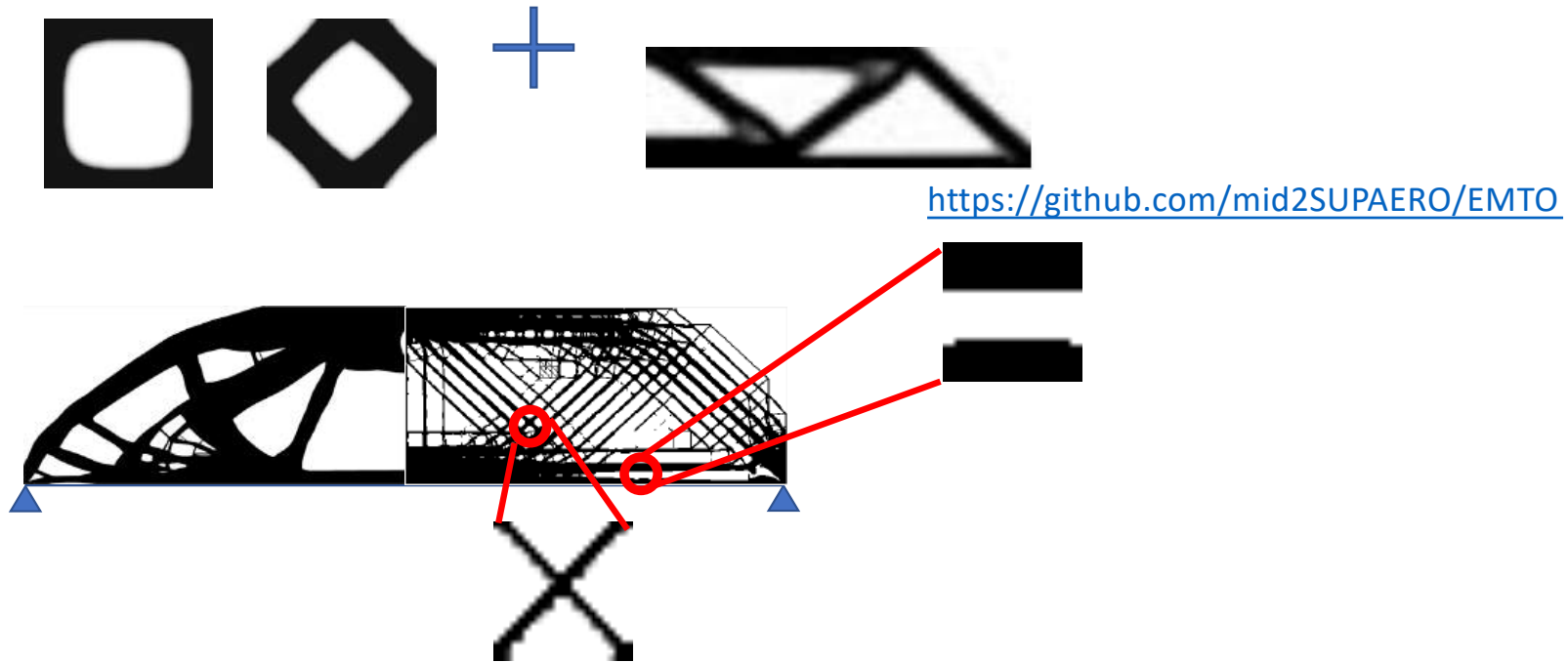


– One step further :
multiscale topology
optimization



Multi-scale TO (well connected+ locally-oriented)

A two level optimization that combines Unit cell design & Topology Optimization



Xia L, Breitkopf P (2015) Design of materials using topology optimization and energy-based homogenization approach in Matlab. Struct Multidisc Optim 52(6):1229–1241.

Wu, Jun, Ole Sigmund, and Jeroen P. Groen. "Topology optimization of multi-scale structures: a review." Structural and Multidisciplinary Optimization 63.3 (2021): 1455-1480.

Multiscale Topology Optimization

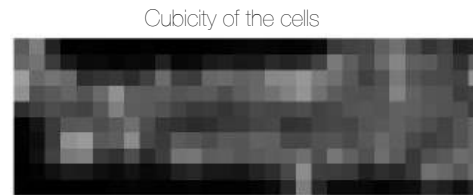
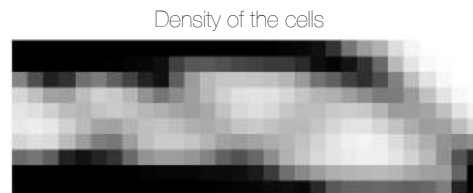
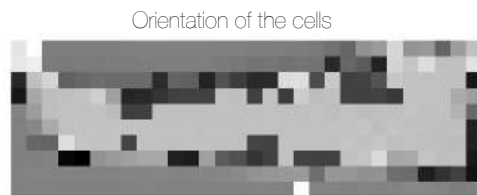
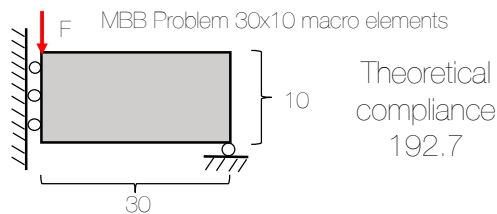
Macroscale Problem need to solve

$$\text{minimize}_{x_{\text{dens}}^i, x_a^i, x_b^i, \dots} c = u^T K u$$

$$\text{subject to } K u = f$$

$$\sum_{i=1}^n \sum_{j=1}^m \rho_{i,j} \leq n \times m \times v_f$$

$$\epsilon < \rho_{i,j} < 1$$



best unit cell per quad

$$x^i = [x_{\text{dens}}^i, x_{\text{or}}^i, x_{\text{cub}}^i]$$

Microscale Problem Unit cell with 3 properties
macro-density, angle, cubicity

$$\text{minimize}_{\rho_{ij}} c_i = E_{1111}^{i\alpha} \times \left(1 - \frac{x_{\text{cub}}^i}{2}\right) + E_{2222}^{i\alpha} \times \frac{x_{\text{cub}}^i}{2}$$

$$\text{subject to } K_i u_i^{A(pq)} = f_i^{(pq)}$$

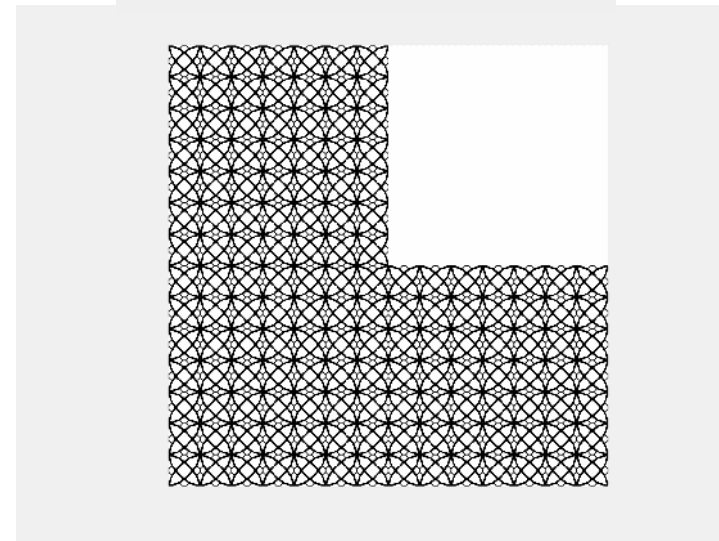
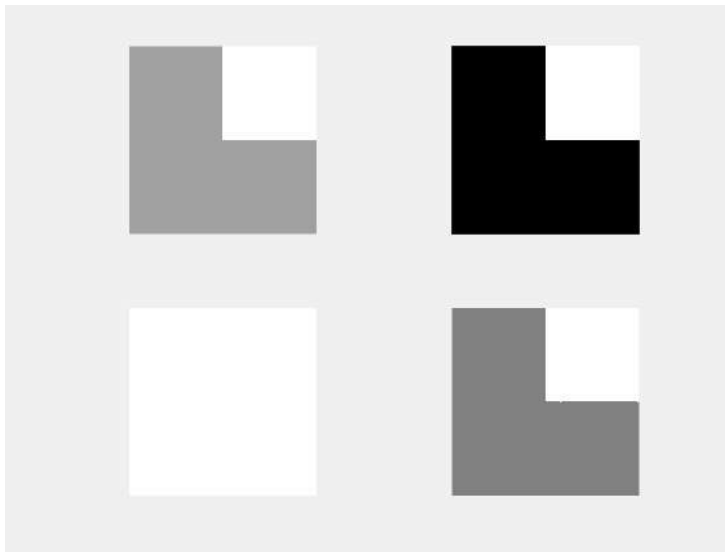
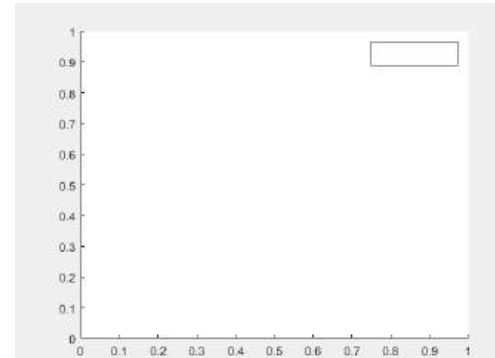
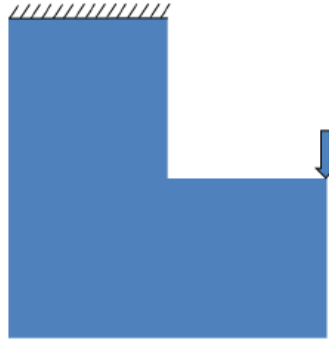
$$\sum_{j=1}^m \rho_{i,j} \leq m \times x_{\text{dens}}^i$$

Since the objective is to create micro-structure with optimal properties towards specific directions, the objective function is a weighted function of the two components $E\alpha_{1111}$ and $E\alpha_{2222}$



Result on classical test cases

- Validation on small grid
⇒ Evaluate full-scale design

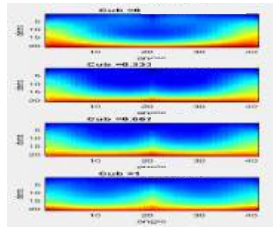


- 4x14*14 design variables; stopping criteria : $\text{tolfun} < 10^{-3}$

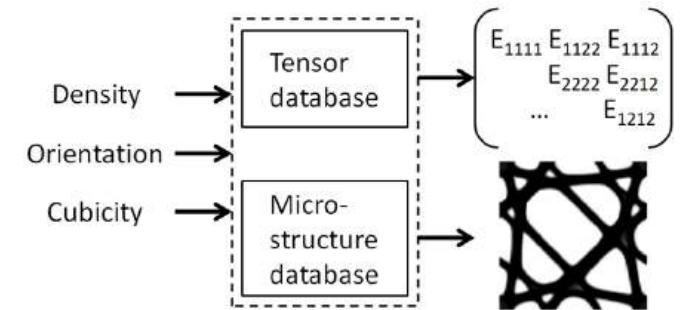
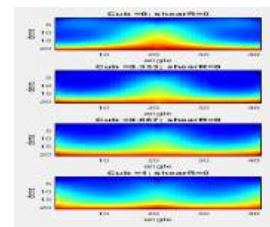
Accelerating Multiscale Topology Optimization

to Avoid Nelement*cellOptimization at each macroOptimization step

- 3 inputs : macro-density, angle, cubicity
- 6 outputs : elastic tensor values
- Gaussian interpolation : capture local effects but mitigate noise
- E1111:



E2222:



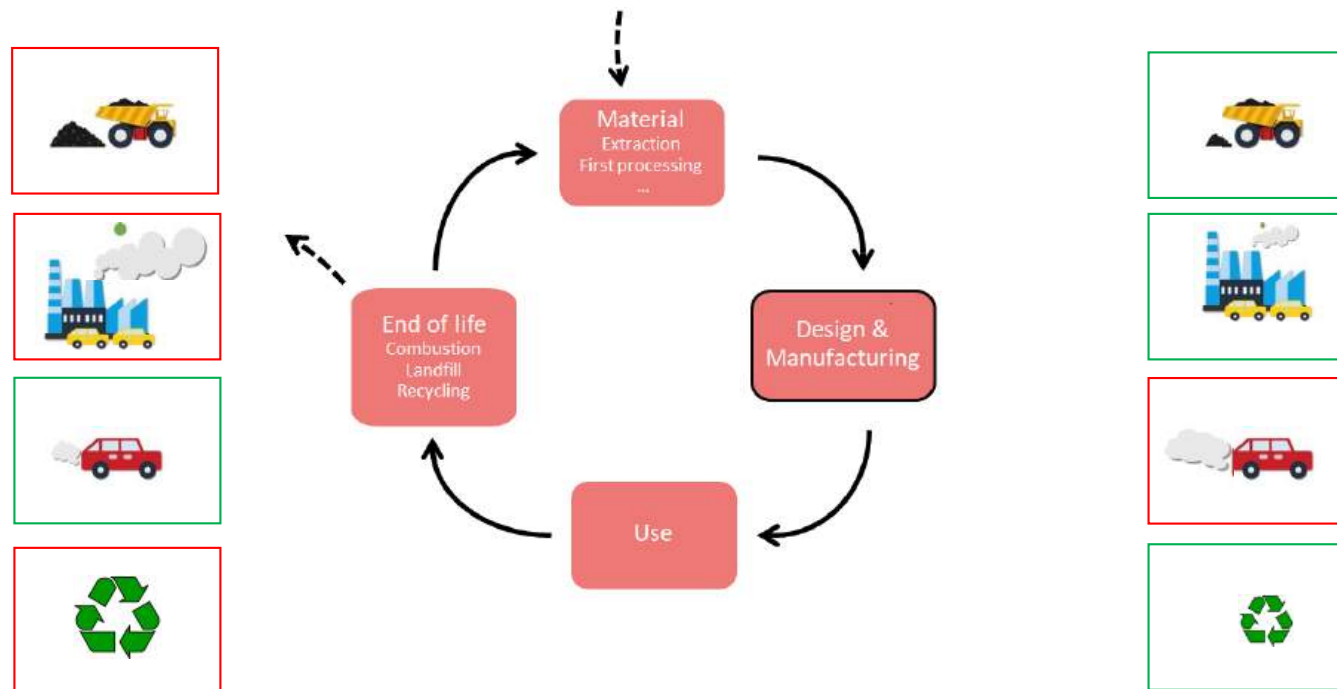
Duriez, Edouard (2021), "EMTO-8tz-database", Mendeley Data, V1, doi: 10.17632/b5hyzvg7fv.1

<https://github.com/mid2SUPAERO/EMTO>

Séminaire Pluridisciplinaire d'Optimisation de Toulouse

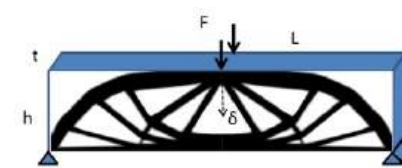
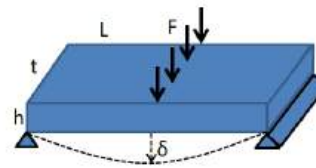
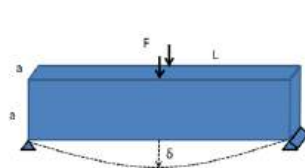
Hypothesis 2

- CO_2 emissions minimization of parts
 - If material choice is **free** => more complicated



First zoom

Missing point from Ashby's theory: The absence of a simple analytical relation between compliance and volume fraction.



Procedia CIRP
Volume 109, 2022, Pages 454–459



Ecodesign with topology optimization

Edouard Duriez ^a, Joseph Morlier ^a, Catherine Azzaro-Pantel ^b, Miguel Charlotte ^a

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<https://doi.org/10.1016/j.procir.2022.05.278>

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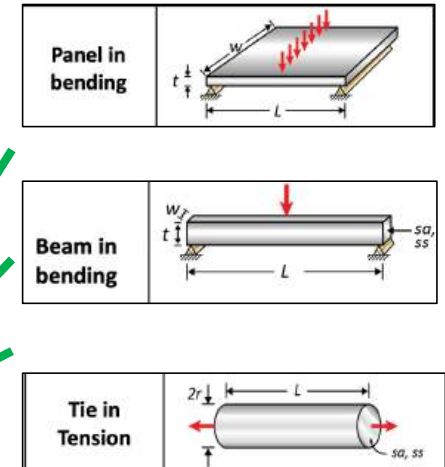
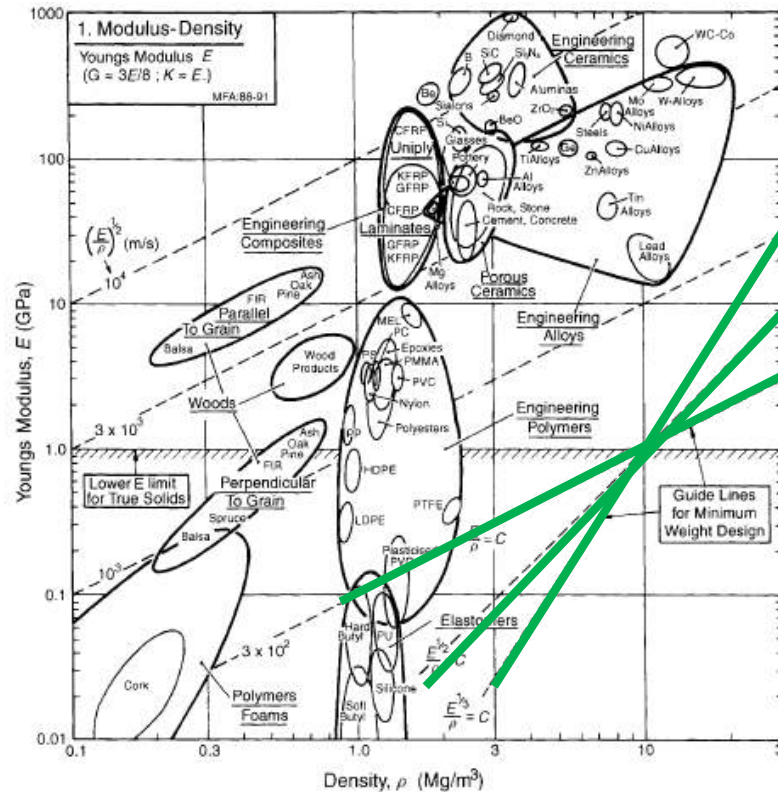
REMINDER !!

- Many materials
 - Competing properties
 - => Ashby indexes: $f_3(M)$
- $$P = f_1(F) \times f_2(G) \times f_3(M)$$

F: Functional constraints

G: Geometrical constraints

M: Material properties



M. F. Ashby et Kara Johnson. « Materials and Design : The Art and Science of Material Selection in Product Design ». In : 2002.

2 examples ... on a n dimensional space

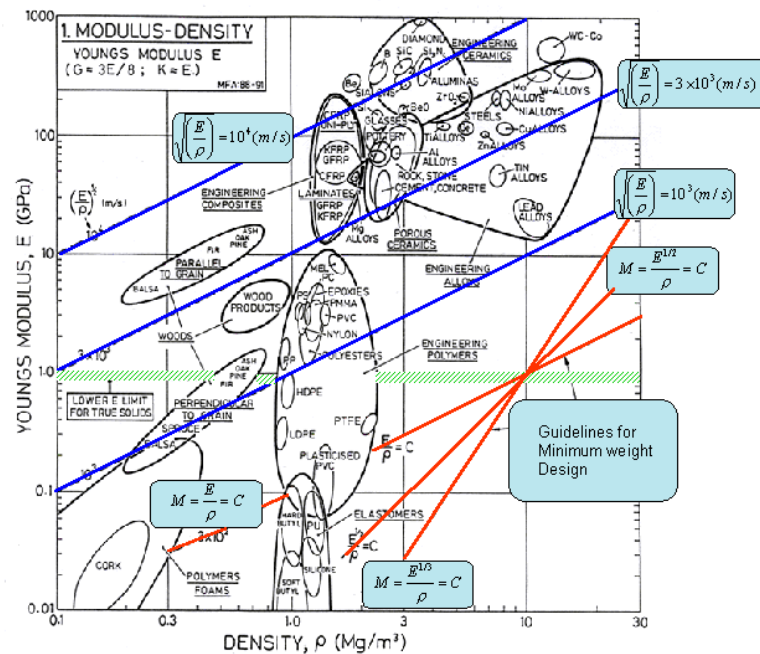


Figure 4-2: The Modulus-Density materials selection chart

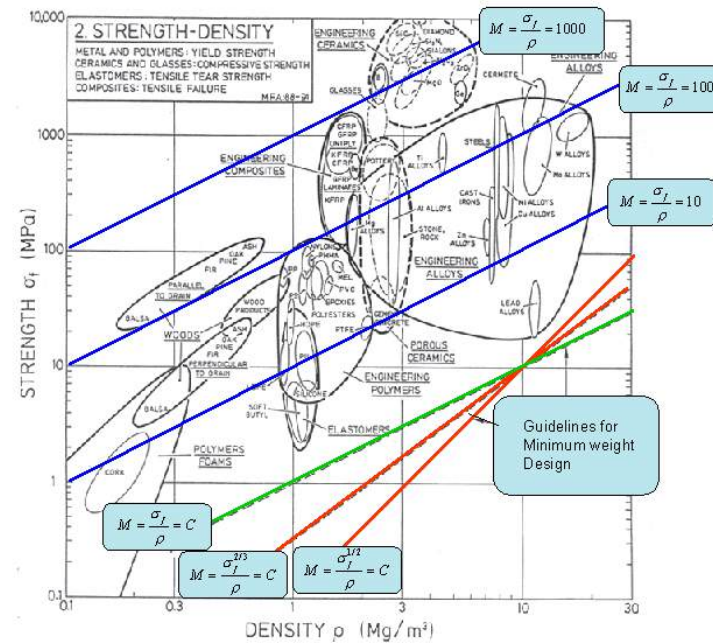
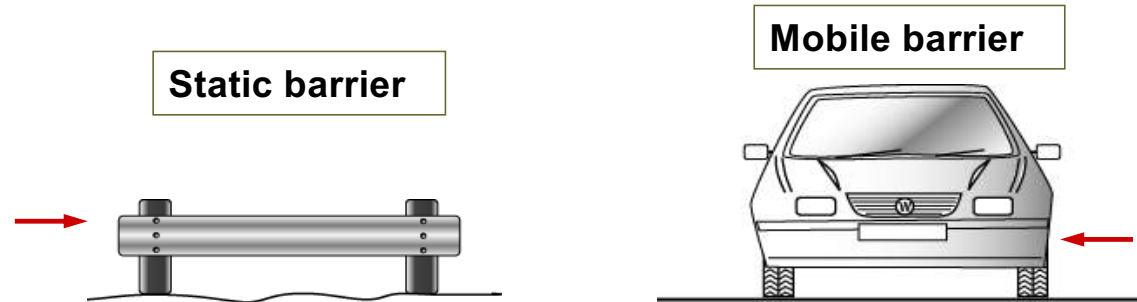


Figure 4-1: The Strength-Density materials selection chart

ME 474-674 Materials Selection for Mechanical Design

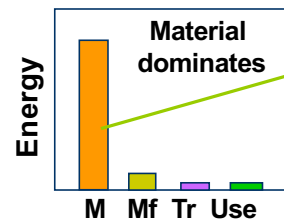
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Same structure, 2 different applications

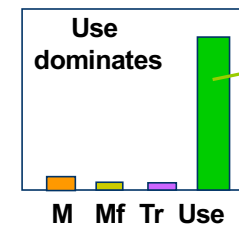


Function: *Absorb impact, transmit load to energy-absorbing units or supports*

Dominant phase of life:



Minimize embodied energy



Minimize mass

Criterion:

Bending strength per unit embodied energy

Bending strength per unit mass

Index:

$$\frac{\sigma_y^{2/3}}{H_m \rho}$$

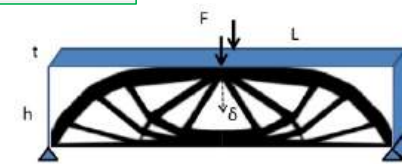
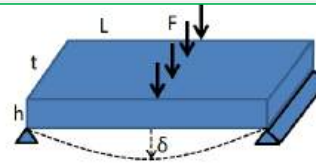
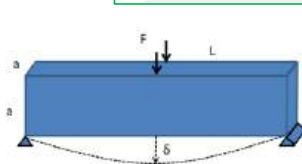
$$\frac{\sigma_y^{2/3}}{\rho}$$

Séminaire Pluridisciplinaire d'Optimisation de Toulouse

Material index

- If fixed material and process,
 CO_2 minimization = mass minimization
- Material choice through indices introduced by Ashby
=> uncouple material choice and part sizing
- Include the geometrical design ($\mathcal{D}$) in the variables :

Properties	Bending beam (Ashby)	Bending plate (Ashby)	Duriez et al. (2022b)
Free variables	a, m	h, m	$t, \mathcal{D}, m$
Fixed	$L, \mathcal{D}$	$L, t, \mathcal{D}$	$L_{\max}, h_{\max}$
Constraint	$\delta_{\max}$	$\delta_{\max}$	$\delta_{\max}$

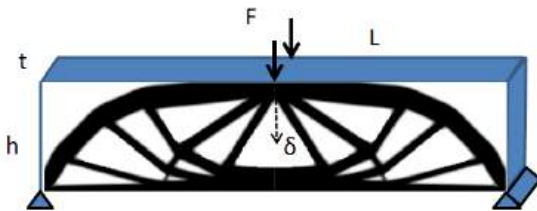


Ashby, M.F., 2004. Materials selection in mechanical design. 2. ed., reprinted ed., Elsevier Butterworth-Heinemann, Amsterdam.

Séminaire Pluridisciplinaire d'Optimisation de Toulouse

Deriving the material index

- Problem considered:



$$\arg \min_{mat, \mathcal{D}, t} CO_2^{tot}(mat, \mathcal{D}, t)$$

$$s.t. \quad \delta \leq \delta_{max}$$

$$mat = \{E, \rho, CO_{2mat}^i\} \in \Phi$$

$$0 < v_f(\mathcal{D}) \leq 1$$

- Objective function:

$$CO_2^{tot} = CO_2^{mat} \times M + CO_2^{veh} \times LD \times M$$

How many miles does an airplane like a 777 fly over the course of its lifetime?

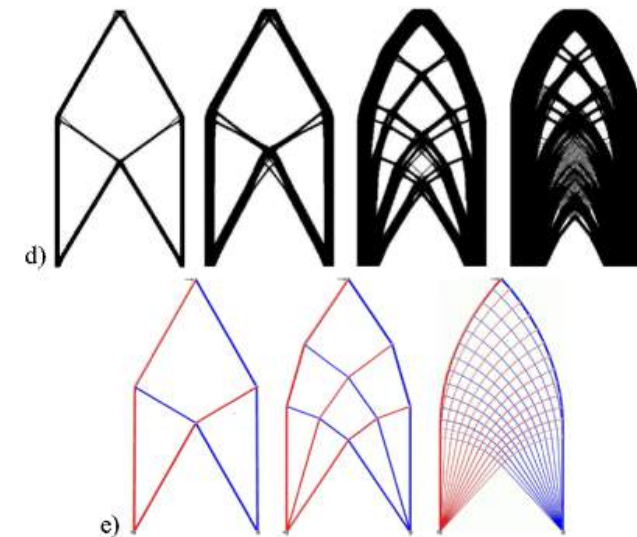
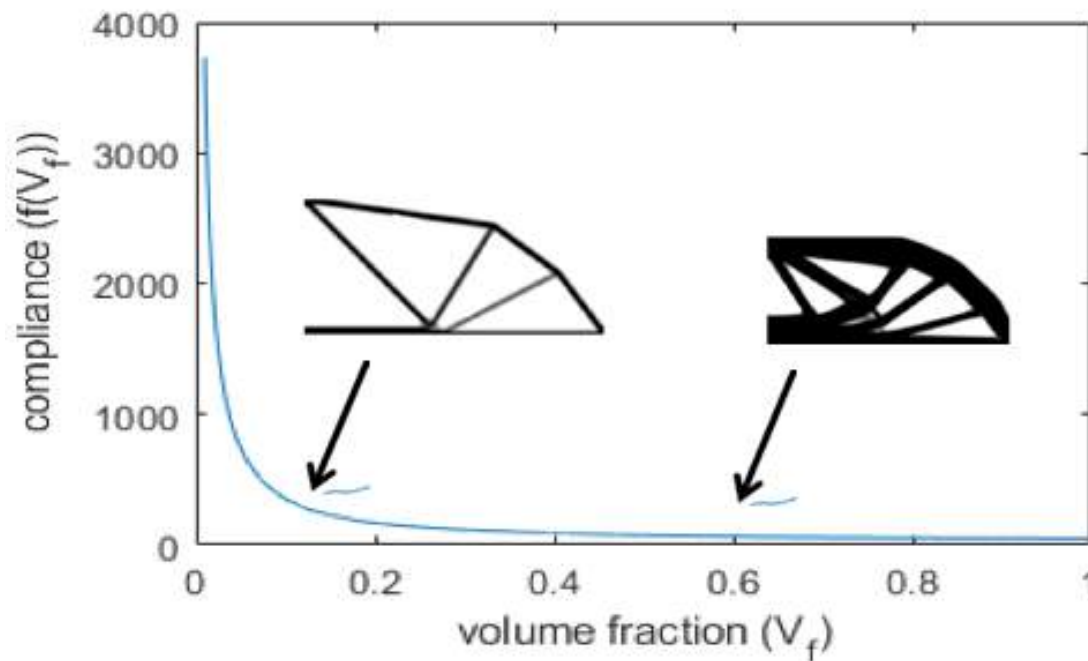
777: A 30-year lifetime. 3,500 hours a year as an average. An average speed of 500 miles per hour. $30 \times 3500 \times 500 = 52,500,000$ miles i.e. LD.

Topology optimization pareto front

V_f : volume fraction (ratio of space containing material)

$f(V_f)$: compliance – volume fraction pareto front

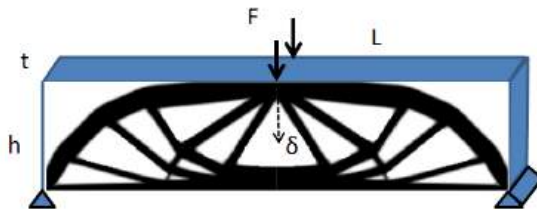
Edouard Duriez, Miguel Charlotte, Catherine Azzaro-Pantel et al. On some properties of the compliance-volume fraction Pareto front in topology optimization useful for material selection., 27 December 2022, PREPRINT (Version 1) available at Research Square [<https://doi.org/10.21203/rs.3.rs-2390440/v1>]



Sigmund, O., Aage, N., & Andreassen, E. (2016). On the (non-) optimality of Michell structures. Structural and Multidisciplinary Optimization, 54, 361-373.

Deriving the material index

- Problem considered:



$$\begin{aligned} \arg \min_{mat, \mathcal{D}, t} \quad & CO_2^{tot}(mat, \mathcal{D}, t) \\ s.t. \quad & \delta \leq \delta_{max} \\ & mat = \{E, \rho, CO_{2mat}^i\} \in \Phi \\ & 0 < v_f(\mathcal{D}) \leq 1 \end{aligned}$$

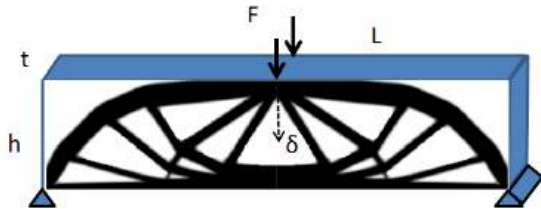
- Objective function:

$$CO_2^{tot} = CO_2^{mat} \times M + CO_2^{veh} \times LD \times M$$

Deriving the material index

starting from $C = U^T K U = U^T F = U(i_F) F(i_F) = F U(i_F)$
 $M = \rho t L h V_f$

- Problem considered:



compliance

$$C \leq F \delta_{max}$$

$$\frac{f(V_f) F}{t E} = \delta_{max}$$

If t is a free variable, it can be chosen as in compliance to achieve the minimum mass

$$t = \frac{f(V_f) F}{\delta_{max} E} \quad \text{thus} \quad M = \frac{L h F}{\delta_{max}} \frac{\rho}{E} f(V_f) V_f$$

$$CO_2^{tot} = \underbrace{(CO_2^{mat} + LD CO_2^{veh})}_{\text{Material } f_3(M)} \times \underbrace{\frac{\rho}{E} \frac{L h F}{\delta_{max}}}_{\text{functional } f_1(F)} \underbrace{f(V_f) V_f}_{\text{topology index } f_2(G)}$$

Material $f_3(M)$

functional $f_1(F)$

topology index $f_2(G)$

Minimize topology index topology index $f_2(G)$

- Topology index : $f(V_f)V_f$:
Minimize this to minimize mass
- Thickness t is finally adjusted to satisfy constraint

On some properties of the compliance-volume fraction Pareto front in topology optimization useful for material selection

Edouard Duriez¹, Miguel Charlotte², Catherine Azzaro-Pantel¹, and Joseph Morlier^{1,2}
¹INSTITUT CLÉMENT ADR, Université de Toulouse, ISAE-SUPAERO, MINES ALBI, UPS, INSA, CNRS, 3 Rue Caroline Aigle, 31400 Toulouse (France)
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 *joseph.morlier@isae-supero.fr

ABSTRACT

Selecting the optimal material for a part designed through topology optimization is a complex problem. The shape and properties of the Pareto front plays an important role in this selection. In this paper we show that the compliance-volume fraction Pareto front of some topology optimization problems is three-dimensionally convex and smooth. The developed efficiency index (EI) can be interpreted as a measure of how much material added to the design and leading to a increased volume fraction, will be efficient in lowering compliance, compared to the mass efficiency of the material already used. A 2-step proof is given. The constraint is a simple compliance constraint and the objective is to minimize the mass efficiency. This last surrogate model has a maximum error of 6.4% with regard to finite element Pareto front, on the different problems tested. The selection of the optimal material is then successfully based on the mass efficiency of a classical topology optimization structure (MBS beam) with a restricted choice of 4 metals (Aluminium, steel, titanium and invar).

Introduction

The topic of this paper is definitely an open problem in structural optimization. We can restate it with a simple question: **How to solve a coupled (material selection/topology optimization) problem?** This one work is based on very few very close references on the state of the art. The introduction will be divided in two parts representing the state of the methodology: topology optimization and material selection. Classical topology optimization consists in optimizing a structure performance by distributing material within a design space with a fixed choice of material [1]. There is a great variety of methods to do this research with: homogenization, Solid Isotropy Material with Penalization (SIMP), evolutionary methods, level set methods, softening Morphable Components (DMC), Generalized Geometric Projection (GGP), among others. These methods are described in more detail in [2]. Let us consider a classical topology optimization framework, formulated as a material distribution problem such as the one in Eq. (1). The design domain Ω is discretized into elements e characterized by their density ρ_e . The compliance C of the design, defined by the static displacements U and the stiffness matrix K , is minimized under a constraint on volume fraction V_f . The design is subject to a load F :

$$\arg \min_{\rho_e} (C(\rho_e)) = U^T K U \quad (1a)$$

$$\text{s.t. } F = K U \quad (1b)$$

$$0 \leq \rho_e \leq 1, \quad \forall e \in \Omega \quad (1c)$$

$$\sum_{e \in \Omega} \rho_e V_e = V_f \quad (1d)$$

$$\text{In this paper, we use the SIMP approach through the well known code optiSLang to solve this problem. In this approach, during the optimization process, each element } e \in \Omega \text{ has an assigned Young modulus } E_e \text{ obeying Eq. (2). We use a penalization value } p > 1 \text{.}$$

$$E_e = \rho_e^p E_0 \quad (2)$$

Here E_0 stands as a reference Young modulus. For illustration, an example of design or topology obtained after the optimization can be seen in Figure 1.

Appendix

A: Proof that the limit of $\alpha(V_f)$ in 0 is 0.
 At very low volume fractions, a good possible design is a truss T^* with very thin members joining the load application points and the fixed boundary conditions of the problem is well posed. Such boundary conditions prevent rigid-body motions. It can be noticed that the topology optimization (SIMP) algorithm automatically self-organizes this truss (elements) so that they can only exist in a truss configuration (i.e. bar behavior). The optimal thin design will have a maximum local proportional thickness to one member, as the total volume fraction varies. As a result, as the volume fraction V_f tends towards 0, the stiffness of the total structure will diminish in proportion with the members' thickness, themselves proportional to the volume fraction. In this case the optimal stiffness κ_{opt} can be written as in Eq. (3) with a specific constant real constant α :

$$\kappa_{opt} = \alpha V_f^{-\alpha}, \quad \alpha > 0 \quad \text{for } V_f > 0 \quad (3)$$

If this case is not the optimal design, the optimal design must be stiffer.

As all the designs for $V_f > 0^+$ are the same with $\alpha(V_f) > 0$ ($\alpha(V_f) > 0$), we have:

$$\frac{\partial \kappa_{opt}}{\partial V_f} (V_f) > \frac{\partial \kappa_{opt}}{\partial V_f} (V_f) > 0 \quad (4)$$

Because $\kappa_{opt}(V_f) > 0$, we can write Eq. (5):

$$\lim_{V_f \rightarrow 0^+} \frac{\kappa_{opt}(V_f)}{V_f} > \lim_{V_f \rightarrow 0^+} \frac{\kappa_{opt}(V_f)}{V_f} > 0 \quad (5)$$

Therefore, we can write Eq. (6):

$$\lim_{V_f \rightarrow 0^+} \frac{\partial \kappa_{opt}}{\partial V_f} (V_f) > \lim_{V_f \rightarrow 0^+} \frac{\partial \kappa_{opt}}{\partial V_f} (V_f) = 1 \quad (6)$$

Proof that α is positive.

Let us prove that κ_{opt} is an increasing function. We proceed by contradiction.

We suppose that κ_{opt} is not an increasing function of V_f . Therefore,

$$\exists V_f \in (0, 1), \exists \Delta V_f > 0 \text{ such that } \kappa_{opt}(V_f) > \kappa_{opt}(V_f + \Delta V_f) \quad (7)$$

We consider the design obtained at the end of the topology optimization for the volume fraction V_f , and the stiffness of which is $\kappa_{opt}(V_f)$. We can increase the density of all elements with $\rho \in [0, 1]$ a same amount such that we obtain a new design with a volume fraction increased by ΔV_f . This new design is stiffer than the initial one, because of Eq. (7). Therefore, we have found a design with a volume fraction of $V_f + \Delta V_f$ and a higher (effective) stiffness (κ_e) than the initial (optimal) one at V_f . This contradicts with Eq. (5).

$$\kappa_e > \kappa_{opt}(V_f) > \kappa_{opt}(V_f + \Delta V_f) \quad (8)$$

However, $\kappa_{opt}(V_f + \Delta V_f)$ is the highest stiffness accessible to a design of volume fraction $V_f + \Delta V_f$, because it is the Pareto front. Therefore, $\kappa_e \leq \kappa_{opt}(V_f + \Delta V_f)$. This is in evident contradiction with Eq. (8) and entails therefore that κ_{opt} is an increasing function of V_f and $\forall V_f \in (0, 1), \kappa_{opt}(V_f) > 0$.

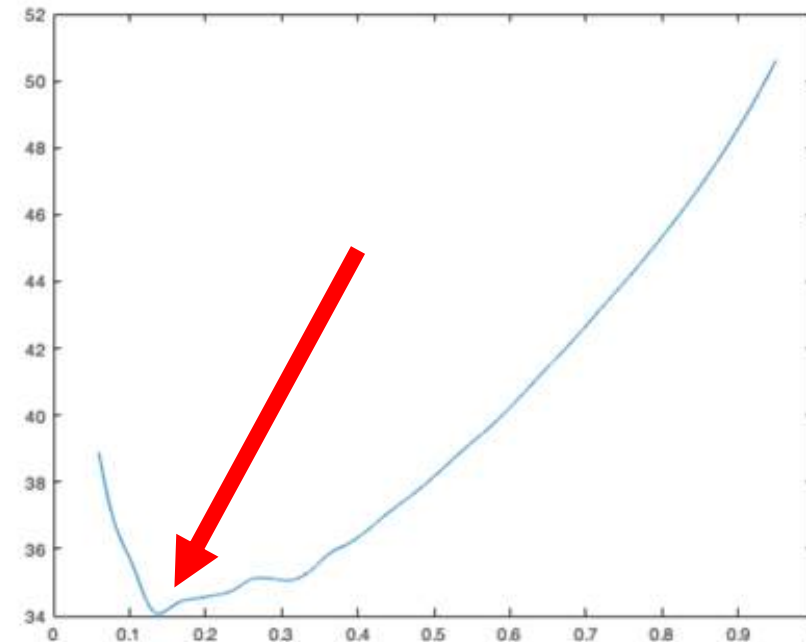
Furthermore, we know that $0 < V_f \leq 1$ and $\kappa_{opt}(V_f) > 0$. Hence, as a conclusion,

$$\frac{\partial \kappa_{opt}}{\partial V_f} (V_f) > 0, \quad \forall V_f \in (0, 1) \quad (9)$$

Proof that $\alpha < 1$.

In this subsection we first prove the above assertion in a particular case where the topology optimization is well posed.

Then, we explain why we assume whether this can be generalized or not.



Data for Material index

functional improvement can be found in lightweight components for transport systems

Table 8. **Fuel** consumption reduction coefficients for different vehicle types and related life time impact savings per kg of weight reduction.

Transport system	Energy source	FRC [26]	Service life	Eco-Impact (ReCiPe H/A)	Life time savings (ReCiPe H/A)	Equivalent electrical energy
Gasoline car	Gasoline	0.5 l / (100kg*100km)	200000km	0.121 Pts/l	1.21 Pts/kg	85 MJ
Diesel car	Diesel	0.24 l / (100kg*100km)	200000km	0.141 Pts/l	0.68 Pts/kg	48 MJ
Short distance train	Electricity	300 kJ / (1000kg*km)	3.5*10 ⁶ km	0.051 Pts/kWh	14.88 Pts/kg	1050 MJ
Long distance train	Electricity	100 kJ / (1000kg*km)	10*10 ⁶ km	0.051 Pts/kWh	14.17 Pts/kg	1000 MJ
Short distance aircraft	Kerosene	12.5 ton / (100kg*year)	25 year	0.134 Pts/l	335 Pts/kg	23647 MJ
Long distance aircraft	Kerosene	103 ton / (100kg*year)	25 year	0.134 Pts/l	2760 Pts/kg	194852 MJ

Kellens, Karel, et al. "Environmental impact of additive manufacturing processes: does AM contribute to a more sustainable way of part manufacturing?." *Procedia Cirp* 61 (2017): 582-587.

Results

- Results change depending on the application:

- Aircraft

Material	E (GPa)	ρ (kg/m ³)	CO_{2mat}^i (kgCO ₂ /kg)	Idx (kgCO ₂ /N/m)
Al alloy	70.8	2795	13.0	3.90×10^{-3}
Stainless steel	197	7915	6.15	3.97×10^{-3}
Ti alloy	115	4425	40.4	3.80×10^{-3}
Inconel 713	205	7900	15.5	3.81×10^{-3}

mass : 5,24kg

5.2445

CO₂ emissions:
517 tons

1.0e+05 * 5.1744

- bridge

Material	E (GPa)	ρ (kg/m ³)	CO_{2mat}^i (kgCO ₂ /kg)	Idx (kgCO ₂ /N/m)
Al alloy	70.8	2795	13.0	5.13×10^{-7}
Stainless steel	197	7915	6.15	2.47×10^{-7}
Ti alloy	115	4425	40.4	1.56×10^{-6}
Inconel 713	205	7900	15.5	5.97×10^{-7}

mass : 5,47kg

5.4761

CO₂ emissions:
33,67 kg

33.6783

To go deeper

Duriez, Edouard (2022), “eco-3dPrinting”, Mendeley Data, V1, doi: 10.17632/rkjddxz3jd.1



Cleaner Environmental Systems

Volume 9, June 2023, 100114



A fast method of material, design and process eco-selection via topology optimization, for additive manufactured structures

Edouard Duriez ^a  , Catherine Azzaro-Pantel ^b, Joseph Morlier ^a, Miguel Charlotte ^a

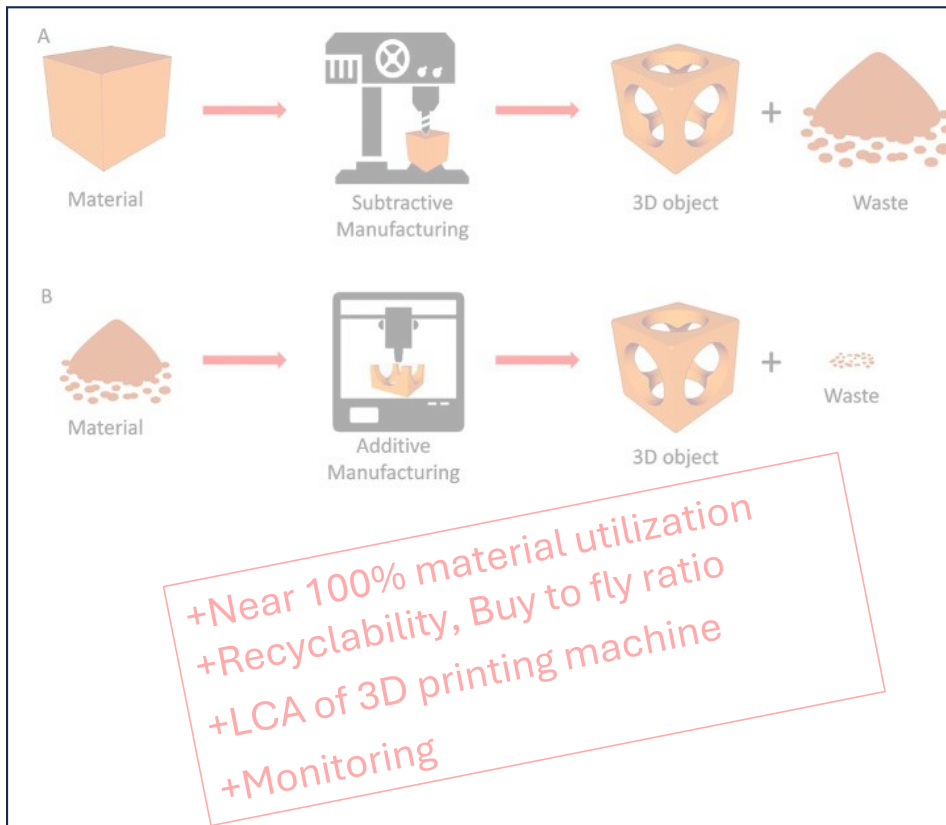
Properties	Bending beam (Ashby)	Bending plate (Ashby)	Duriez et al. (2022b)	Our problem
Free variables	a, m	h, m	$t, \mathcal{D}, m$	$\mathcal{D}, m, p$
Fixed	$L, \mathcal{D}$	$L, t, \mathcal{D}$	$L_{\max}, h_{\max}$	$L_{\max}, h_{\max}, t_{\max}$
Constraint	$\delta_{\max}$	$\delta_{\max}$	$\delta_{\max}$	$\delta_{\max}$

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Few years later

Process is AM, but WHY?

<https://dfam.substack.com/p/dfam-education-in-2022>

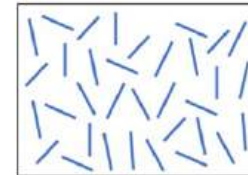


Regular and periodic



Natural (optimal?)

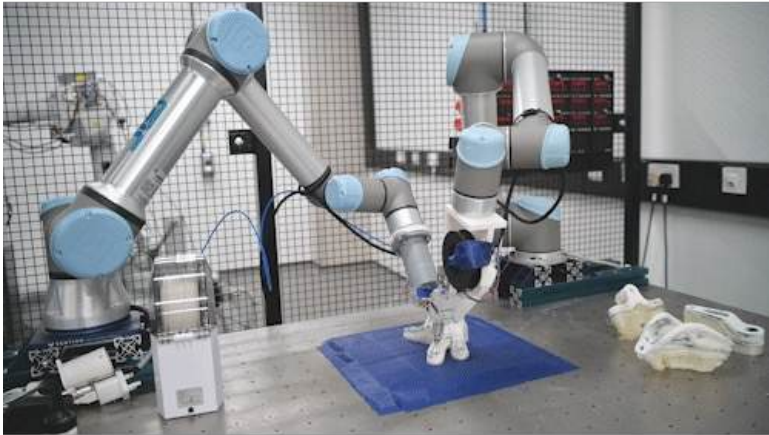
Random



Non-periodic and specific (optimal)

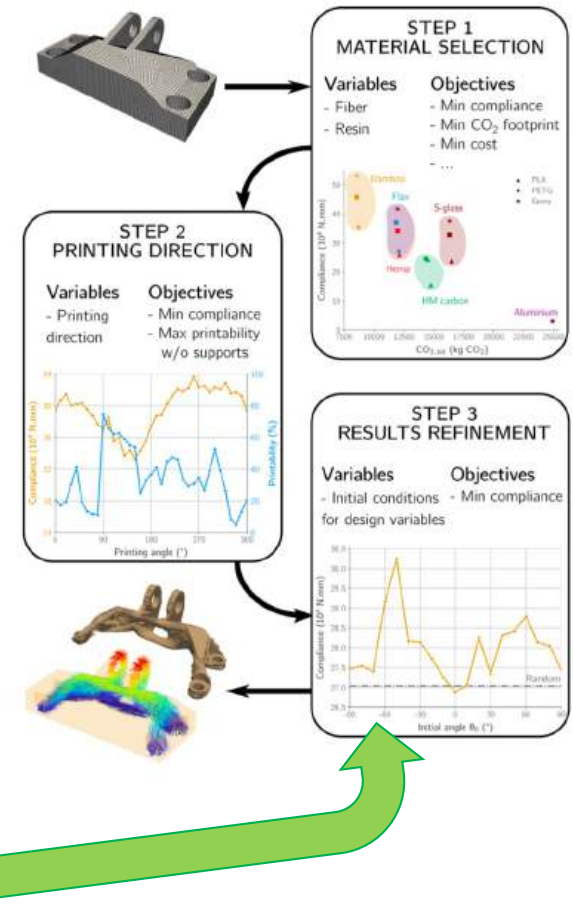
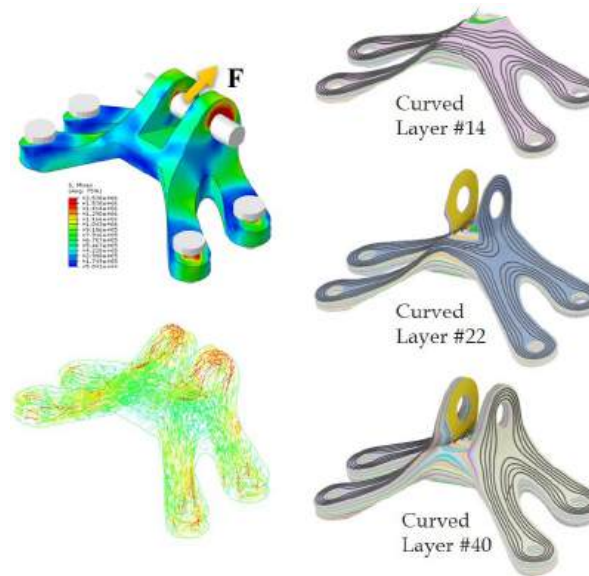
+ Automatic Fiber Placement + eco-fiber/resin selection +Monitoring

Ongoing work <https://www.youtube.com/watch?v=7Jxyu9uRMLo>



Fang, G., Zhang, T., Huang, Y., Zhang, Z., Masania, K., & Wang, C. C. (2024). Exceptional mechanical performance by spatial printing with continuous fiber: Curved slicing, toolpath generation and physical verification. *Additive Manufacturing*, 104048.

https://github.com/mid2SUPAERO/SOMP_Ansys



Asai, G., Jansari, C., Lachaud, F., Masania, K., & Morlier, J. (2025). Ecodesign of 3D volumetric fiber-composite structures with topology optimization. *Composites Part A: Applied Science and Manufacturing*, 190, 108615.

Add Ecomaterial selection, printability

Results



Composites Part A: Applied Science and
Manufacturing
Volume 190, March 2025, 108615



Ecodesign of 3D volumetric fiber-composite structures with topology optimization

Gustavo Asai ^{a, b}, Chintan Jansari ^d, Frédéric Lachaud ^c, Kunal Masania ^d, Joseph Morlier ^c  

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https://github.com/mid2SUPAERO/SOMP_Ansys

Au programme

1. Problem $\mathcal{P}_1$, where the topology is fixed (truss) or parametric (wing),
2. Problem $\mathcal{P}_2$, where the topology is free.

3. On some coupled problems $\mathcal{P}_3$

4. Conclusions

Aerospace design is a coupled problem $\mathcal{P}_3$

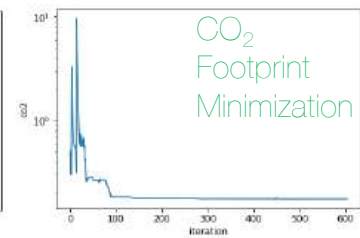
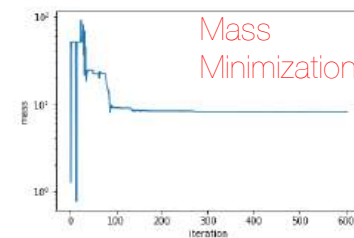
- MDO implies to solve the coupling of physics (aerodynamic and structures)
- To limit the computational cost of high-fidelity simulations we extensively use surrogate modeling and bayesian optimization

Aircraft Design Fluid x control x physics x applied maths x structures & materials

In AD: Strong coupling
Between
Disciplines (MDO)

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
		Avionics group	Electrical group	Escape system	Armament	Landing gear	Hydraulics group	Flight control system	Environment and control system	Power plant group	Fatigue group	Aero elastic group	Stress group	Materials group	Fly wheel	Empennage group	Rear fuselage	Fuselage group
Avionics group	1	●	●	●	●	●	●	●	●	●						●	●	●
Electrical group	2	●	●	●	●	●	●	●	●	●						●	●	●
Escape system	3	●	●	●	●	●	●	●	●	●						●	●	●
Armament	4	●	●	●	●	●	●	●	●	●	●	●	●	●		●	●	●
Landing gear	5	●	●	●	●	●	●	●	●	●	●	●	●	●		●	●	●
Hydraulics group	6	●	●	●	●	●	●	●	●	●	●	●	●	●		●	●	●
Flight control system	7	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Environment and control system	8	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Power plant group	9	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Fatigue group	10	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Aero elastic group	11	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Stress group	12	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Materials group	13	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Empennage group	14	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Wing group	15	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Rear fuselage	16	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Fuselage group	17	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

Topology x Material x Process



$$\text{Range} = V t_f = V \times \underbrace{\left(\frac{L}{D}\right)}_{\text{aircraft designer}} \times \underbrace{I_{sp}}_{\text{propulsion system designer}} \times \underbrace{\ln\left(\frac{W_i}{W_f}\right)}_{\text{structural designer}}$$

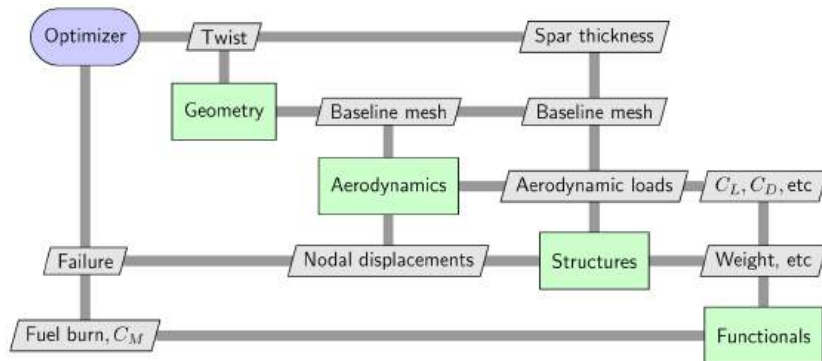
MDO

needs derivative (coupled), gradient based, scalable, need a good x_0

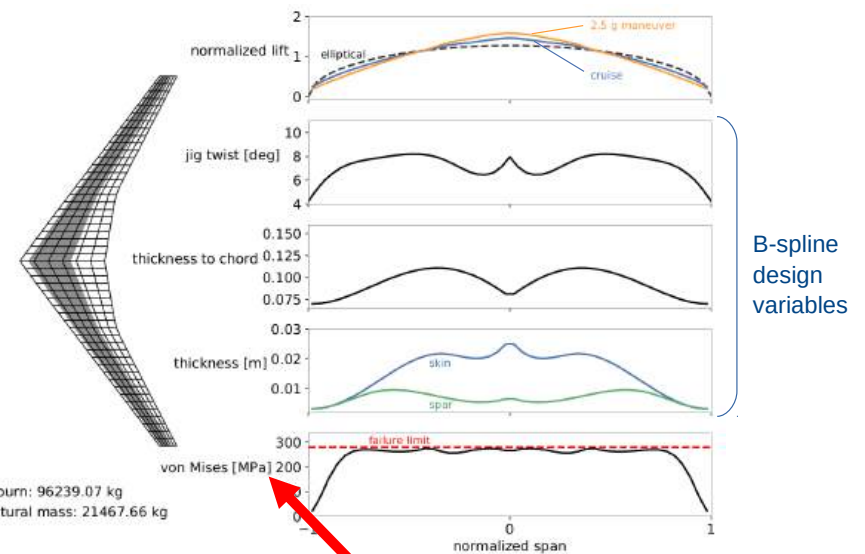
Results

This is MDO

Discipline Structure is coupled to
Discipline Aerodynamic !



Minimize fuel burn



Wrt

L=W

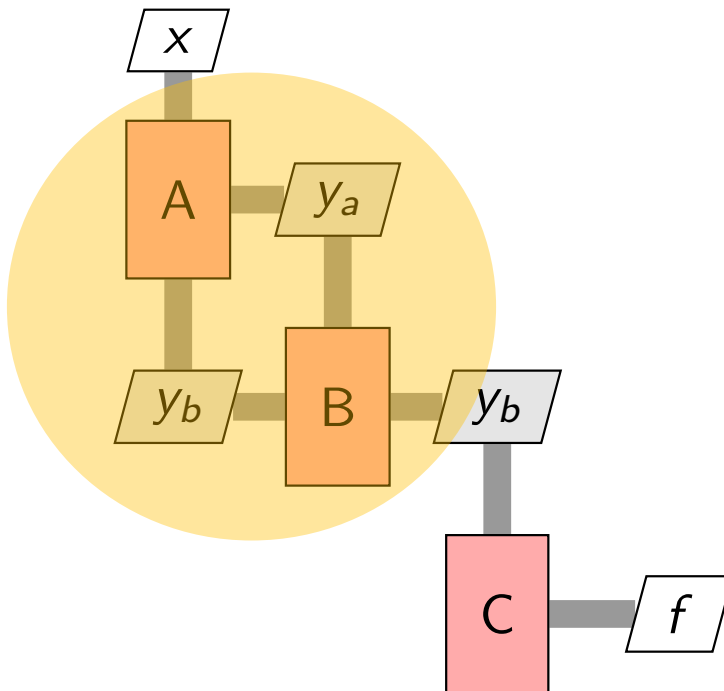
Admissible stress

Etc...

Jasa, J. P., Hwang, J. T., & Martins, J. R. (2018). Open-source coupled aerostructural optimization using Python. *Structural and Multidisciplinary Optimization*, 57(4), 1815-1827.

Chauhan, S. S., & Martins, J. R. (2018, September). Low-fidelity aerostructural optimization of aircraft wings with a simplified wingbox model using OpenAeroStruct. In *International conference on engineering optimization* (pp. 418-431). Cham: Springer International SGR 27/11/25

Tricky part ... coupled adjoint



The coupled system is defined by:

$$y_a(y_b) = 2 + 2 \cdot y_b + x^2 \quad (A)$$

$$y_b(y_a) = 0.25 \cdot y_a - 0.25 \quad (B)$$

$$f = 2 \cdot y_b \quad (C)$$

Compute the total derivative

$$\frac{df}{dx} = \frac{\partial f}{\partial y_b} \frac{\partial y_b}{\partial y_a} \frac{\partial y_a}{\partial x} + \frac{\partial f}{\partial y_b} \frac{dy_b}{dx} + \frac{\partial f}{\partial y_a} \frac{dy_a}{dx}$$

MDO toolkit (opensource)

Frameworks

1

OpenMDAO



A computing platform for solving MDO problems that automates the complex data passing and derivative calculations.

2

CasADi



A symbolic framework for numeric optimization, particularly strong for problems that can be formulated with Automatic Differentiation.

GEMSEO

<https://gemseo.readthedocs.io/en/stable/>

Optimizers (Solvers)

3

IPOPT (Interior Point Optimizer)



A powerful solver for large-scale nonlinear optimization, often used with adjoint-based methods.

4

SLSQP (Sequential Least Squares Quadratic Programming)



A versatile gradient-based optimizer that effectively handles constraints.

5

MMA (Method of Moving Asymptotes)



A popular and effective solver for large-scale topology optimization problems.

README Apache-2.0 license



EGObbox - Efficient Global Optimization toolbox

tests passing pytest passing lint passing JOSS 10.21505/joss.04737

Rust toolbox for Efficient Global Optimization inspired by [the EGO implementation](#) in the [SMT](#) Python library.

The `egobox` package is twofold:

1. for end-users: [a Python module](#), the Python binding of the optimizer named `Egoz` and the surrogate model `Gpx`, mixture of Gaussian processes, written in Rust.
2. for developers: [a set of Rust libraries](#) useful to implement bayesian optimization (EGO-like) algorithms,

<https://github.com/reif/egobox>



<https://smt.readthedocs.io/en/latest/>

Testcase #1: Aircraft optimization

Topology is free in the MDO problem

f1=Minimize (-range) and f2=minimise (GWP) % second problem

$$f = \alpha * f1 + (1 - \alpha) * f2$$



<https://pymoo.org>



LCA4MDAO

<https://github.com/mid2SUPAERO/LCA4MDAO>

Hybrid Aircraft Problem (MDOLab)

- Hybridised King Air C90GT from [OpenConcept](#), built in *OpenMDAO* format
- Four disciplines:
 - Aero (wing geometry)
 - Propulsion (with hybrid system)
 - Structure
 - Trajectory simulation
- 6 variables converted into LCA database entries

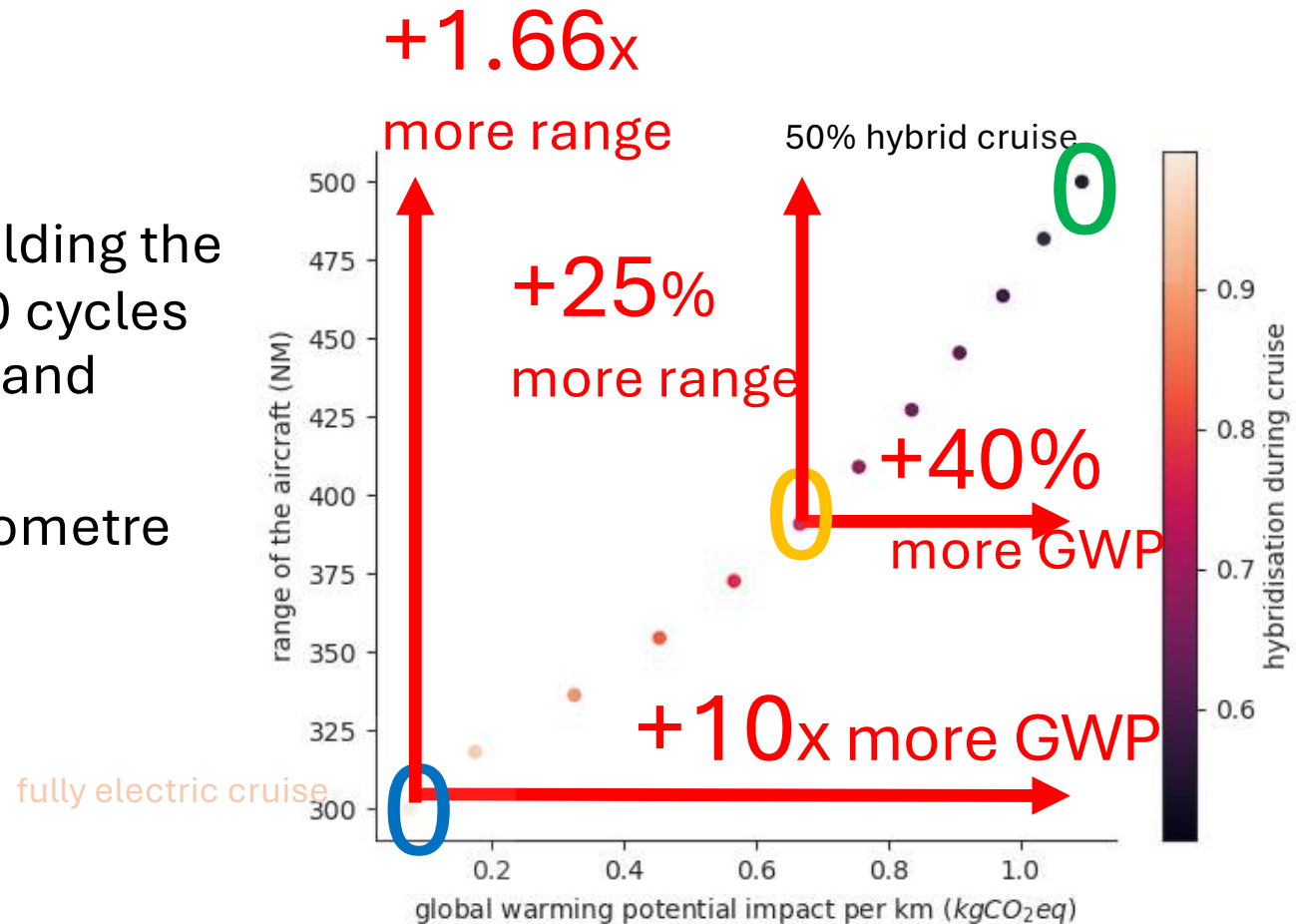
Model parameter	Ecoinvent entry
Battery weight	battery cell production, Li-ion
Motor weight	electric motor production, vehicle
Engine weight	internal combustion engine production, passenger car
Empty weight	aluminium production, primary, ingot
Fuel used	market for kerosene
Electricity used	market group for electricity, low voltage

Benjamin J. Brelje and Joaquim R. R. A. Martins, "Development of a Conceptual Design Model for Aircraft Electric Propulsion with Efficient Gradients", 2018 AIAA/IEEE Electric Aircraft Technologies Symposium, AIAA Propulsion and Energy Forum, (AIAA 2018-4979) DOI: 10.2514/6.2018-4979

Eytan J. Adler and Joaquim R. R. A. Martins, "Efficient Aerostructural Wing Optimization Considering Mission Analysis", Journal of Aircraft, 2022. DOI: 10.2514/1.c037096

Results MOO

- LCA scope include building the aircraft and flying 1000 cycles at max range with fuel and electricity
- Functional unit is a kilometre flown

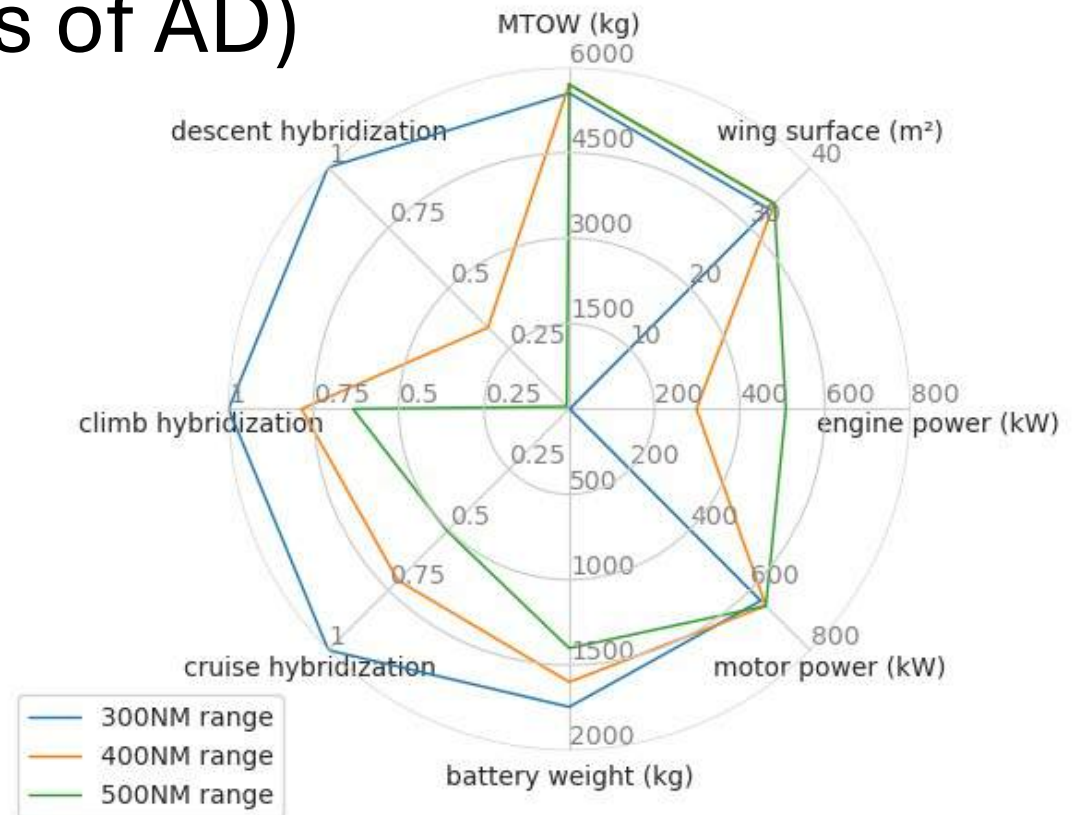


Results (link to physics of AD)

For the design variables,
reducing the range:

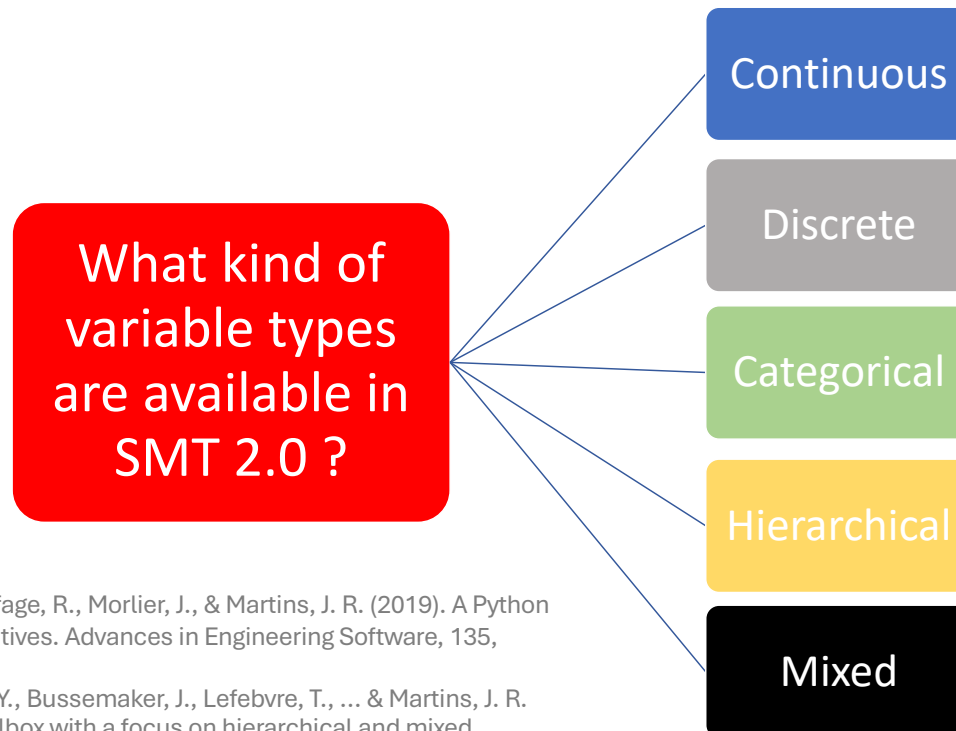
- increases the hybridization
- reduces the engine size
- increases the battery weight

variable	value	units
MTOW	5700	kg
wing surface	34	m ²
engine power	298	kW
motor power	652	kW
battery weight	1607	kg
fuel capacity	500	kg
cruise hybridisation	0.71	
climb hybridisation	0.785	
descent hybridisation	0.337	
GWP	0.712	kgCO ₂ eq/km



Another solution... Another story

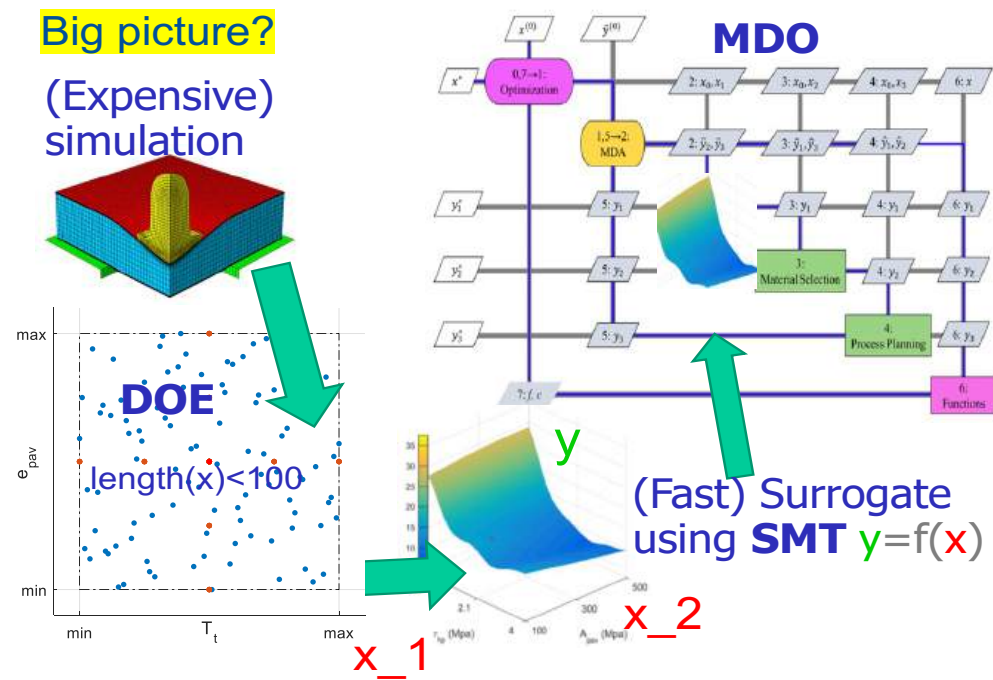
- *material selection as hybrid/categorical variables ?*



Bouhlef, M. A., Hwang, J. T., Bartoli, N., Lafage, R., Morlier, J., & Martins, J. R. (2019). A Python surrogate modeling framework with derivatives. *Advances in Engineering Software*, 135, 102662.

Saves, P., Lafage, R., Bartoli, N., Diouane, Y., Bussemaker, J., Lefebvre, T., ... & Martins, J. R. (2024). SMT 2.0: A Surrogate Modeling Toolbox with a focus on hierarchical and mixed variables Gaussian processes. *Advances in Engineering Software*, 188, 103571.

Interested in Surrogate modeling? GP/BO?



<https://github.com/SMTorg/smt>

Au programme

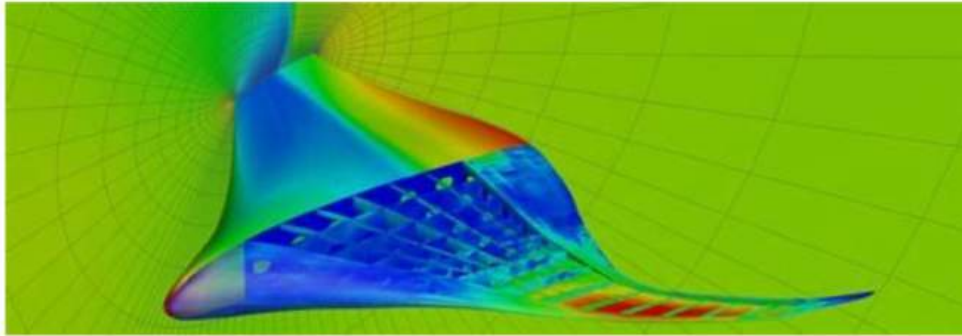
1. Problem $\mathcal{P}_1$, where the topology is fixed (truss) or imposed (wing),
2. Problem $\mathcal{P}_2$, where the topology is free.
3. On some coupled problems $\mathcal{P}_3$

4. Conclusions

Conclusions

- Proof of concept of greener structures
- VAE, Surrogate Optimization can help for Design acceleration especially when material choice is in the loop
- Need help for writing proofs in some of my papers 😊





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<http://mdolab.engin.umich.edu>

Optimization [MDO] for connecting people?

Publié le 14 février 2019 [Modifier l'article](#) | [Voir les stats](#)



joseph morlier

Professor in Structural and Multidisciplinary Design Optimization, ... any idea?

2 articles



74



31



3



0



https://www.tripadvisor.fr/LocationPhotoDirectLink-g187529-d574612-i349532022-Museum_of_Natural_Science_Museo_de_Ciencias_Naturales-Valencia_Province_o.html

Is it possible to build an aircraft wing in LEGO® ?

Publié le 17 février 2020 [Modifier l'article](#) | [Voir les stats](#)



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Professor in Structural and Multidisciplinary Design Optimization, ... any idea?

5 articles

<https://www.linkedin.com/pulse/possible-build-aircraft-wing-lego-joseph-morlier/?articleId=6627240732975480832>

<https://www.linkedin.com/pulse/optimization-mdo-connecting-people-joseph-morlier/>